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Schedule of the Day

Section 21
Monitoring, measurement, analysis, and evaluation

Section 22
Internal audit

Section 23
Management review

Section 24
Treatment of nonconformities

Section 25
Continual improvement

Section 26
Preparing for the certification audit

Section 27
Closing of the training course

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By the end of the day, participants will be able to:

- Monitor, measure, analyze, and evaluate the effectiveness of the ISMS
- Establish ISMS performance indicators
- Establish an internal audit program, plan and perform audit activities, and follow up on nonconformities
- Prepare, conduct, and close a management review
- Treat nonconformities
- Maintain and continually improve the ISMS
- Prepare for the certification audit, stage 1 and stage 2 audit, and audit follow-up

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Section 21

Monitoring, measurement, analysis, and evaluation

- Determine measurement objectives
- Define what needs to be monitored and measured
- Establish ISMS performance indicators
- Determine the frequency and method of monitoring and measurement
- Report the results

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This section aims at providing participants with information on how to determine the measurement objectives, define what aspects of an ISMS need to be monitored and measured, and establish performance indicators. Various methods of reporting the measurement results will be given, and the participant will be able to use the acquired knowledge and skills to verify the extent to which the identified ISO/IEC 27001 requirements have been met.

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Monitoring, Measurement, Analysis, and Evaluation

1. Define and establish		2. Implement and operate		3. Monitor and review		4. Maintain and improve	
1.1	Understanding the organization and its context	2.1	Selection and design of controls	3.1	**Monitoring, measurement, analysis, and evaluation**	4.1	Treatment of nonconformities
1.2	ISMS scope	2.2	Implementation of controls	3.2	Internal audit	4.2	Continual improvement
1.3	Leadership and project approval	2.3	Management of documented information	3.3	Management review		
1.4	Organizational structure	2.4	Communication				
1.5	Analysis of the existing system	2.5	Competence and awareness				
1.6	Information security policy	2.6	Management of security operations				
1.7	Risk management						
1.8	Statement of Applicability						

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ISO/IEC 27001's Requirements for Monitoring and Measurement

ISO/IEC 27001, clause 9.1

The organization shall evaluate the information security performance and the effectiveness of the information security management system.

The organization shall determine:

a) *what needs to be monitored and measured, including information security processes and controls;*
b) *the methods for monitoring, measurement, analysis and evaluation, as applicable, to ensure valid results. The methods selected should produce comparable and reproducible results to be considered valid;*
c) *when the monitoring and measuring shall be performed;*
d) *who shall monitor and measure;*
e) *when the results from monitoring and measurement shall be analyzed and evaluated;*
f) *who shall analyze and evaluate these results.*

Documented information shall be available as evidence of the results.

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ISO/IEC 27003, clause 9.1 Monitoring, measurement, analysis and evaluation

A good practice is to define the 'information need' when planning the monitoring, measurement, analysis and evaluation. An information need is usually expressed as a high level information security question or statement that helps the organization evaluate information security performance and ISMS effectiveness. In other words, monitoring and measurement should be undertaken to achieve a defined information need.

Care should be taken when determining the attributes to be measured. It is impracticable, costly and counterproductive to measure too many, or the wrong attributes. Besides the costs of measuring, analysing and evaluating numerous attributes, there is a possibility that key issues could be obscured or missed altogether.

There are two generic types of measurements:

h.*performance measurements,* *which express the planned results in terms of the characteristics of the planned activity, such as head counts, milestone accomplishment, or the degree to which information security controls are implemented; and*

i.*effectiveness measurements,* *which express the effect that realization of the planned activities has on the organization's information security objectives.*

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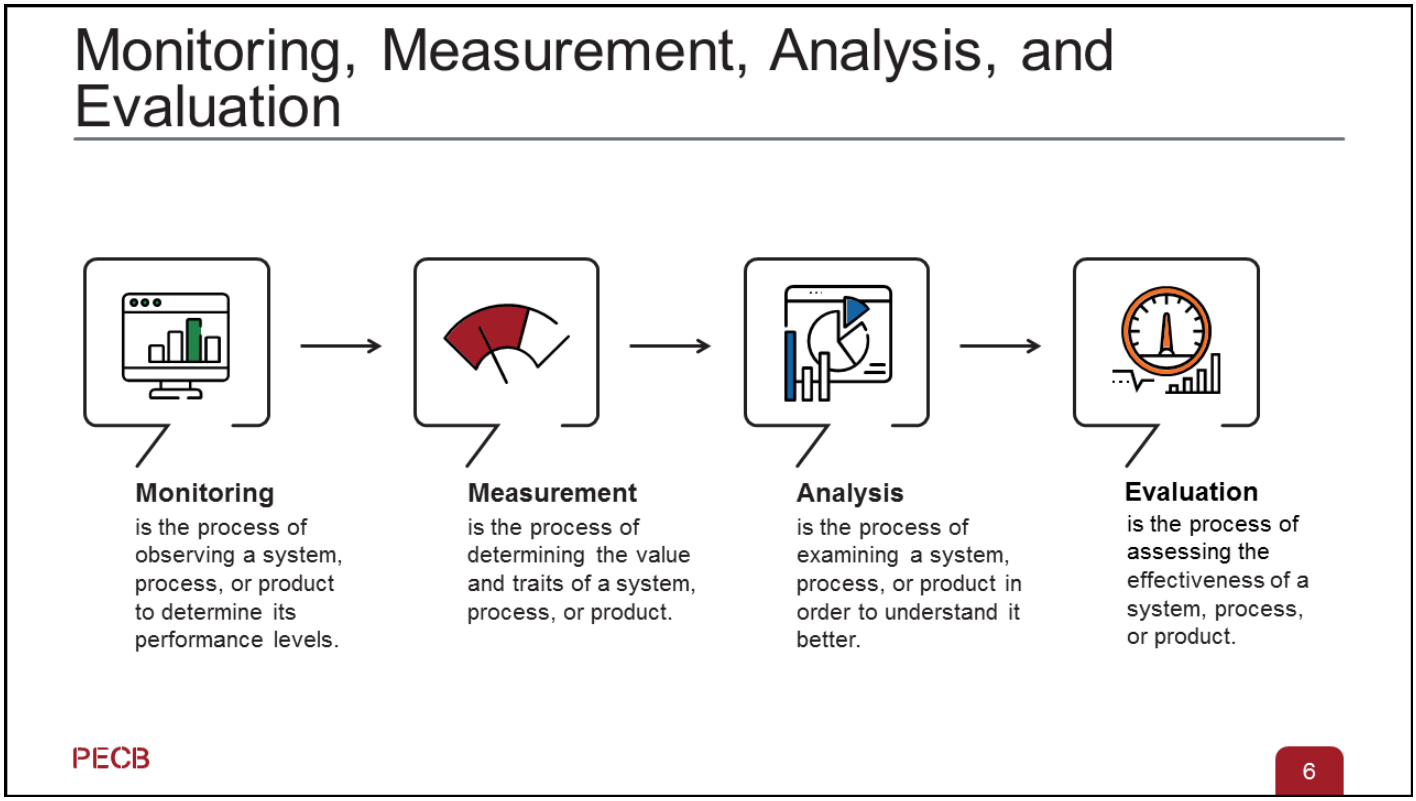

Monitoring, measurement, analysis, and evaluation are critical factors in the assessment of the performance of the ISMS. The goal is to determine the extent to which the processes meet the objectives. The outputs of these activities allow the organization to compare the actual performance levels with desired performance.

Some of the advantages of monitoring, measurement, analysis, and evaluation are:

- Determining the effectiveness and efficiency of processes
- Identifying deviations in a timely manner and treating them accordingly
- Implementing controls to ensure the realization of processes
- Identifying opportunities for continual improvement

Performance measures should be a high priority on the agenda of individuals or a team responsible for the implementation and maintenance of the information security management system.

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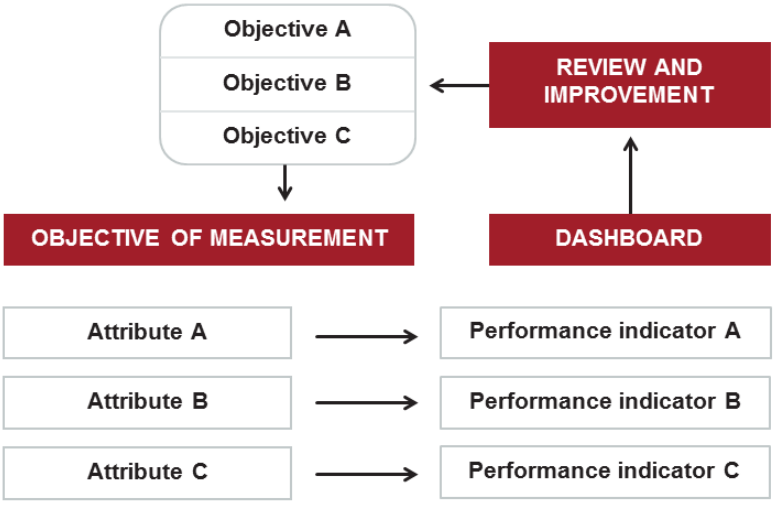
Monitoring, Measurement, Analysis, and Evaluation

The main goal of monitoring, measurement, analysis, and evaluation is to gain insights about the performance of the ISMS.



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In a broader perspective, monitoring and measurement involves:

- Setting objectives for measuring and monitoring
- Selecting the attributes to be measured
- Establishing performance indicators
- Evaluating if the objectives are achieved

Example:

1. **Measurement objectives:** verify if all employees are aware of the major risks that the organization is facing
2. **Attribute:** employee that has attended the awareness session
3. **Performance indicator:** the percentage of employees that have attended the awareness session

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ISO/IEC 27004

- ISO/IEC 27004 provides guidelines to help organizations effectively evaluate ISMS performance in accordance with the requirements of clause 9.1 *Monitoring, measurement, analysis and evaluation* of ISO/IEC 27001.
- It elaborates on monitoring and measuring, establishing procedures, analyzing results, and reviewing and improving monitoring, measurement analysis, and evaluation processes.
- Organizations cannot obtain certification against this standard.

INTERNATIONAL STANDARD

ISO/IEC 27004

Information technology — Security techniques — Information security management — Monitoring, measurement, analysis and evaluation

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ISO/IEC 27004, Introduction

This document is intended to assist organizations to evaluate the information security performance and the effectiveness of an information security management system in order to fulfill the requirements of ISO/IEC 27001:2013, 9.1: monitoring, measurement, analysis and evaluation.

The results of monitoring and measurement of an information security management system (ISMS) can be supportive of decisions relating to ISMS governance, management, operational effectiveness and continual improvement.

As with other ISO/IEC 27000 documents, this document should be considered, interpreted and adapted to suit each organization's specific situation. The concepts and approaches are intended to be broadly applicable but the particular measures that any particular organization requires depend on contextual factors (such as its size, sector, maturity, information security risks, compliance obligations and management style) that vary widely in practice.

This document is recommended for organizations implementing an ISMS that meets the requirements of ISO/IEC 27001. However, it does not establish any new requirements for ISMS which conform to ISO/IEC 27001 or impose any obligations upon organizations to observe the guidelines presented.

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3.1 Monitoring, Measurement, Analysis, and Evaluation

List of activities

- **3.1.1** Set measurement objectives
- **3.1.2** Determine what needs to be monitored and measured
- **3.1.3** Determine who will monitor, measure, analyze, and evaluate
- **3.1.4** Establish performance indicators
- **3.1.5** Select methods and determine the frequency
- **3.1.6** Report the results

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3.1.1 Set Measurement Objectives

- The organization should evaluate its ISMS to ensure its continual suitability, adequacy, and effectiveness.
- Organizations should focus on monitoring and measuring activities linked to critical processes that facilitate the achievement of information security objectives.
- Too many measures can distort an organization's focus and blur what is truly important.

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The chosen performance measures will be the means of communicating the success or failure of the ISMS. Any set of performance measures needs to be carefully selected to ensure that they are measuring the route to achieving the organization's objectives.

The objectives of measurement in the context of an ISMS include:

- Evaluating the effectiveness of the processes, procedures, and controls in place
- Verifying the extent to which ISO/IEC 27001 requirements have been met
- Facilitating performance improvement
- Providing input for management review to facilitate decision-making and justify the needed improvements of the ISMS in place

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3.1.2 Decide What Needs to Be Monitored and Measured

- The extent to which the organization's information security policy is followed and objectives are met
- The organization's critical processes, procedures, and functions
- Historical evidence of poor ISMS performance (e.g., nonconformities, near misses, false alarms, failures, incidents)
- Compliance with applicable legal and regulatory requirements, industry best practices, and organization's information security policy and objectives
- Data and results of monitoring and measurement sufficient to facilitate subsequent corrective action analysis

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It is preferable to use a minimum number of effective performance measures that are related to the organization's objectives.

Each set of generic measures will only be effective for some organizations and are adequate for organizations in similar environments. The final mix of measures will be a product of operational, legislative, and cultural context.

Performance measurement levels range from strategic high-level measures to more specific operational or program-level measures. The organization needs to measure only activities crucial for its operation and save time and resources on measuring activities simply because they can be measured. In terms of efficiency, the organization needs meaningful measures to indicate what is happening so that it can decide to either let an activity continue or intervene to take corrective action. In terms of effectiveness, the organization needs measures to understand if the management system is aligned with the organization's needs and objectives.

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Decide What Needs to be Monitored and Measured

ISO/IEC 27004, clause 6.2

Systems, processes and activities which can be monitored include, but are not limited to:

a) *implementation of ISMS processes;*
b) *incident management;*
c) *vulnerability management;*
d) *configuration management;*
e) *security awareness and training;*
f) *access control, firewall and other event logging;*
g) *audit;*
h) *risk assessment process;*
i) *risk treatment process;*
j) *third party risk management;*
k) *business continuity management;*
l) *physical and environmental security management; and*
m) *system monitoring.*

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ISO/IEC 27004, clause 6.2 What to monitor (cont'd)

These monitoring activities produce data (event logs, user interviews, training statistics, incident information, etc.) that can be used to support other measures. In the process of defining attributes to be measured, additional monitoring can be required to provide supporting information.

Note that monitoring can allow an organization to determine whether a risk has materialized, and thereby indicate what action it can take to treat such a risk itself. Note also that there can be certain types of information security controls that have the explicit purpose of monitoring. When using outputs of such controls to support measurement, organizations should ensure that the measurement process takes into account whether the data used was obtained before or after any treatment action was taken.

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Decide What Needs to be Monitored and Measured

ISO/IEC 27004, clause 6.3

ISMS processes and activities that are candidates for measurement include:

a) planning;
b) leadership;
c) risk management;
d) policy management;
e) resource management;
f) communicating;
g) management review;
h) documenting; and
i) auditing.

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ISO/IEC 27004, clause 6.3 What to measure (cont'd)

With regards to information security performance, the most obvious candidates are the organization's information security controls or groups of such controls (or even the entire risk treatment plan). These controls are determined through the process of risk treatment and are referred to in ISO/IEC 27001 as necessary controls. They can be ISO/IEC 27001:2013, Annex A controls, sector-specific controls (e.g. as defined in standards such as ISO/IEC 27010), controls specified by other standards and controls that have been designed by the organization. As the purpose of a control is to modify risk, there are a variety of attributes that can be measured, such as:

j.*the degree to which a control reduces the likelihood of the occurrence of an event;*

k.*the degree to which a control reduces the consequence of an event;*

l.*the frequency of events that a control can cope with before failure; and*

m.*how long after the occurrence of an event does it take for the control to detect that the event has occurred.*

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3.1.3 Decide Who Will Monitor, Measure, Analyze, and Evaluate

ISO/IEC 27004, clause 6.5

Whether the measurement is performed manually or automatically, organizations can define the following measurement-related roles and responsibilities:

a) *measurement client: the management or other interested parties requesting or requiring information about the effectiveness of an ISMS, controls or group of controls;*

b) *measurement planner: the person or organizational unit that defines the measurement constructs that links measurable attributes to a specified information need;*

c) *measurement reviewer: the person or organizational unit that validates that the developed measurement constructs are appropriate for evaluating information security performance and the effectiveness of an ISMS, controls or group of controls;*

d) *information owner: the person or organizational unit that owns the information that provides input into measures. This person is responsible for providing the data and is also frequently (but not always) responsible for conducting measurement activities;*

e) *information collector: the person or organizational unit responsible for collecting, recording and storing the data;*

f) *information analyst: the person or organizational unit responsible for analysing data; and*

g) *information communicator: the person or organizational unit responsible for communicating the results of analysis.*

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ISO/IEC 27004, clause 6.5 Who will monitor, measure, analyse and evaluate (cont'd)

Organizations can combine some, or possibly all, of these roles.

Individuals performing different roles and responsibilities throughout the processes can require diverse skill sets and associated awareness and training.

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3.1.4 Establish Performance Indicators

Examples

- The number of information security incidents
- The average cost of an information security incident

Information security cases

- The percentage of employees who have received training regarding information security
- The number of training hours per employee

Training

- The percentage of systems tested for vulnerabilities in the last three months
- The number of days to close the detected vulnerabilities

ISMS weaknesses

- The number of nonconformities detected
- The average number of days required to resolve a nonconformity

Nonconformities

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The types and amount of performance measurements depend on the organization's requirements.

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3.1.5 Select Methods and Determine the Frequency

How and when to monitor and measure?

How?

Practices

- ISO/IEC 27001 does not indicate how, nor how often, must monitoring and measurement be conducted.
- It is up to the organization to determine the methods and frequency of monitoring and measurement.
- It is best practice to use dashboards to record and report on monitoring and measurement activities with performance indicators.
- Dashboards should indicate actual performance vs. predetermined performance targets.

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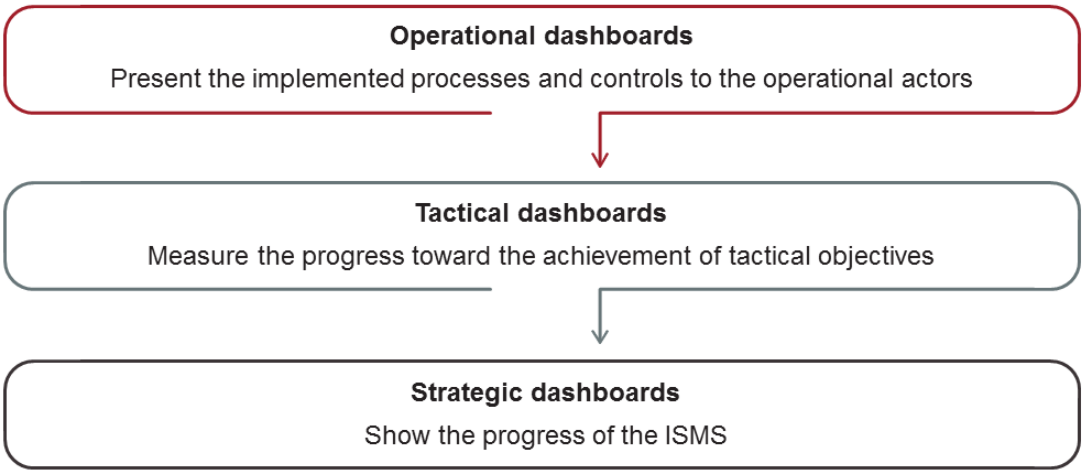
3.1.6 Report the Results

Examples of dashboards



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There are many ways to report the monitoring and measurement results. The selection of the methods will depend on the target audience. The main methods include:

- **Tactical and operational dashboards** are less focused on strategic objectives and more tied to the effectiveness of specific controls or processes.
- **Scorecards or strategic dashboards** provide strategic information by integrating high-level indicators.
- **Reports** (from simple and static in nature, such as a list of measures for a given period, to more sophisticated cross-tab reports with nested grouping, rolling summaries, and dynamic drill-through or linking) are best used when the user needs to look at raw data in an easy-to-read format.
- **Gages** represent dynamic values including alerts, additional graphical elements, and labeling of endpoints.

Note: A dashboard is the user interface that organizes and presents information in a way that is easy to read and understand.

- The dashboard is only the presentation format.
- The indicators are the content.

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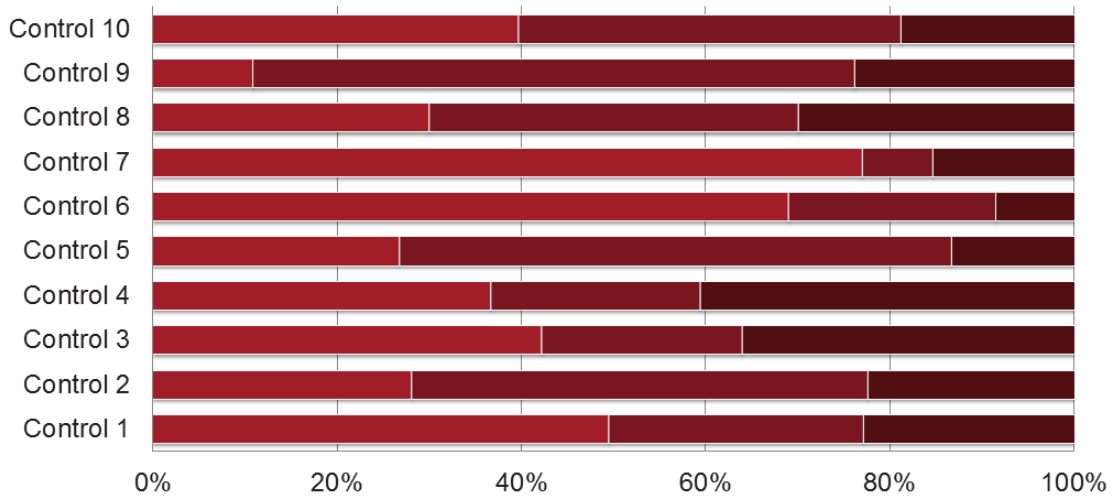
I. Operational Dashboard

Example



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Operational dashboards are used to monitor operations in real time and to notify users about deviations. Furthermore, they help in controlling operational activities and ensuring that processes stay within the targets of productivity, quality, and efficiency. They can assist in analyzing operational performance continuously so as to avoid problems and losses and, at the same time, seize opportunities, while providing data that will help improve process control and efficiency.

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II. Tactical Dashboard

Example

Evaluation of procedures											
No.	Procedure evaluated	Notes to weaknesses and strengths	Level of compliance								
			1	2	3	4	5	6	7	8	9
1	Policy communication									X	
2	Planning of changes								X		
3	Resource allocation								X		
4	Competence, awareness, and training				X						
5	Control of documented information										X
6	Monitoring and measurement								X		
7	Corrective action					X					
8	Management review						X				
9	Internal audits								X		
10	Continual improvement									X	
Overall assessment									X		

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The example on the slide presents the evaluation of the conformity of the procedures related to the management system in a tactical dashboard.

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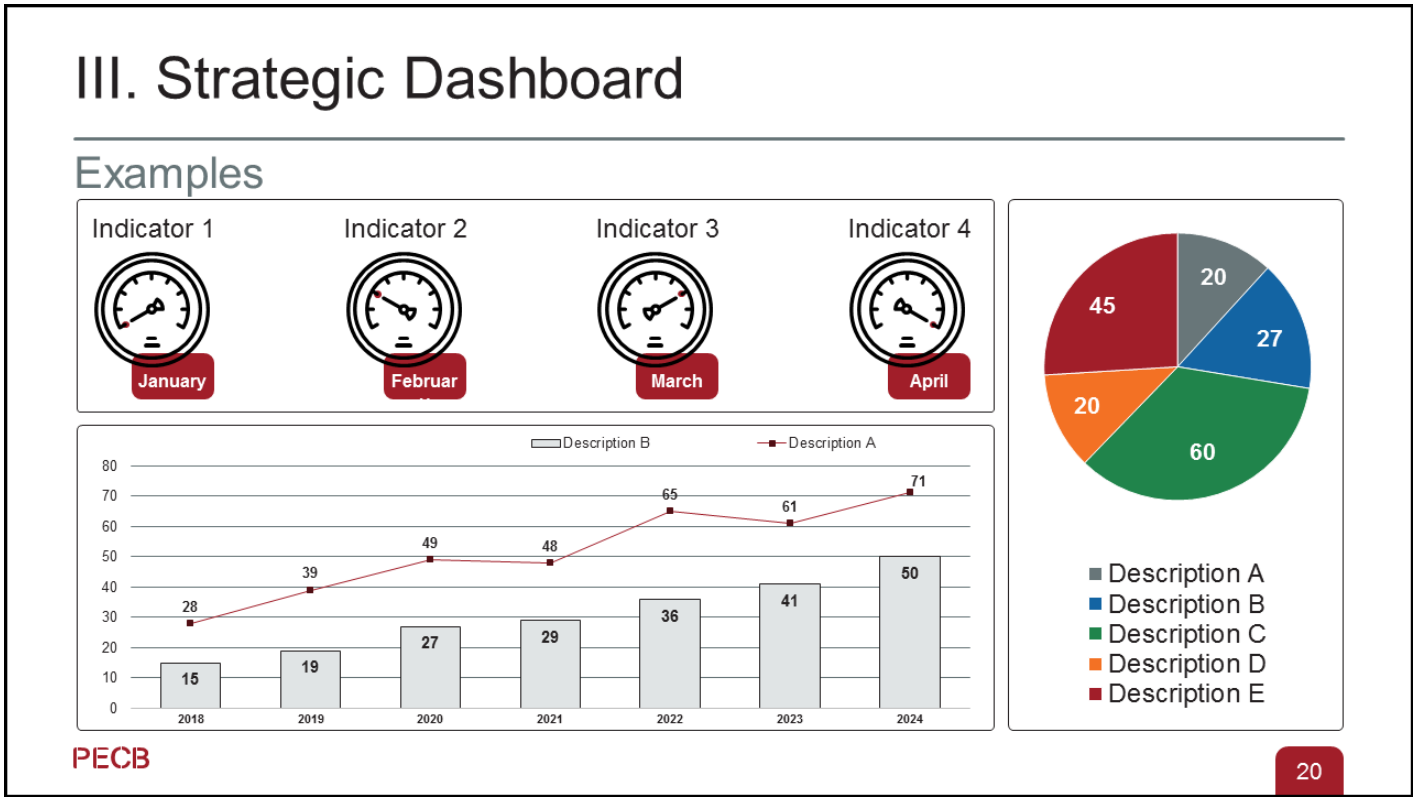

Strategic dashboards are tools that support managers at any level in an organization and provide a quick overview that decision-makers need to monitor the financial health of the business. Dashboards of this type focus on high-level measures of performance and forecasts.

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Section Summary:

- Organizations must evaluate the performance and effectiveness of the ISMS.
- Organizations should identify measurement objectives, establish performance indicators, and evaluate whether the objectives have been met.
- The organization should determine how and how often to monitor or measure the ISMS.
- Results can be reported through methods such as tactical and operational dashboards, scorecards or strategic dashboards, reports, or gages.

Quiz 22

Exercise 4

Questions?

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Note: To complete Quiz 22 and Exercise 4, please go to the Quizzes Sheet and Exercises Sheet respectively.

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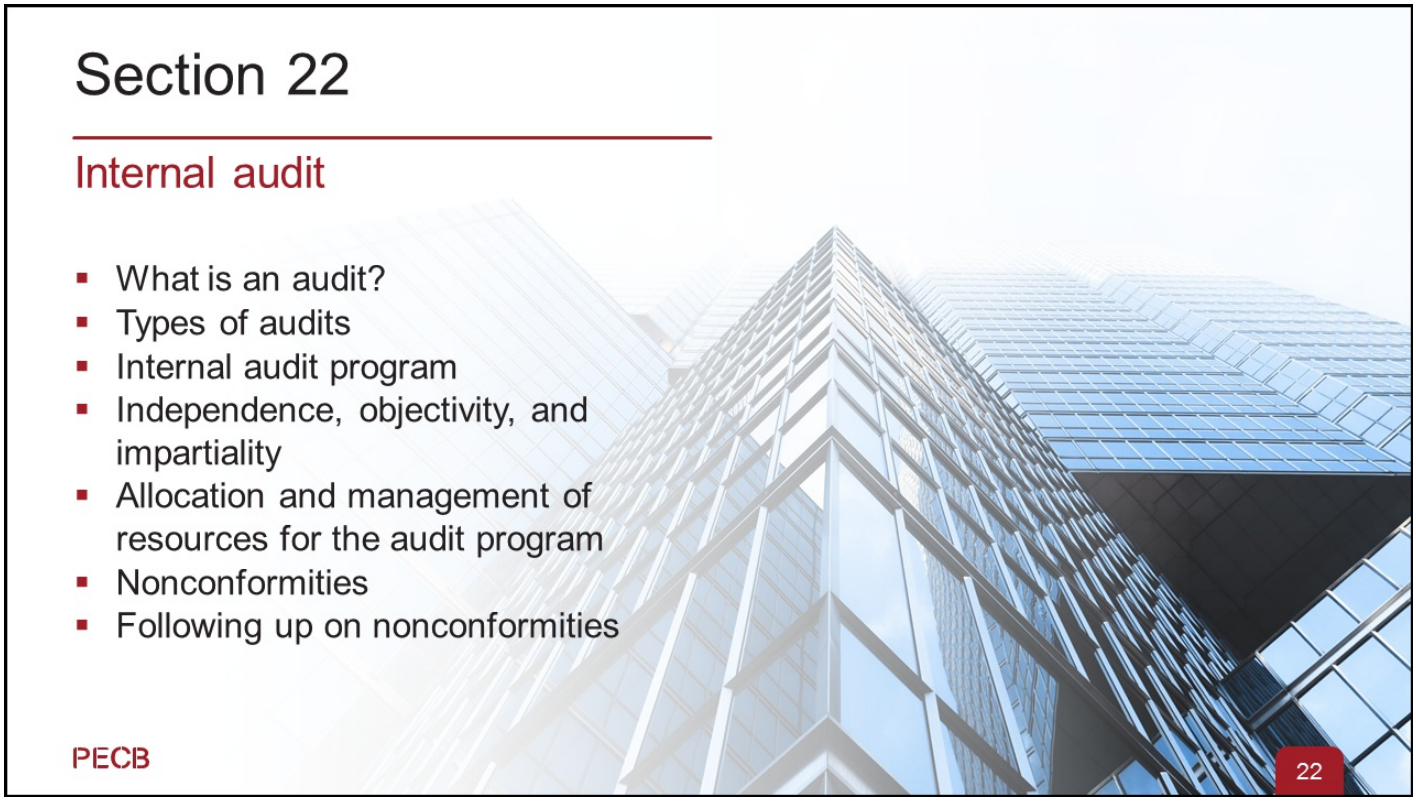

Although the information in this section is applied to internal auditing, the elements of an audit program are essentially the same for first, second, or third party audits. The tools, procedures, and techniques are essentially the same.

The activities performed in internal audits can vary depending on the program objectives, the degree of independence of auditors, the roles of auditors, and business planning. When such differences are significant, additional explanations have been provided in the notes of this section.

This section aims to provide insights into the importance of audit for organizations and the differences between internal and external audits. The section further elaborates on the planning and carrying out of audit activities, and the allocation and management of resources for the audit program. An important factor in this regard is detecting nonconformities as part of the audit and following up on them.

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Internal Audit

1. Define and establish		2. Implement and operate		3. Monitor and review		4. Maintain and improve	
1.1	Understanding the organization and its context	2.1	Selection and design of controls	3.1	Monitoring, measurement, analysis, and evaluation	4.1	Treatment of nonconformities
1.2	ISMS scope	2.2	Implementation of controls	3.2	**Internal audit**	4.2	Continual improvement
1.3	Leadership and project approval	2.3	Management of documented information	3.3	Management review		
1.4	Organizational structure	2.4	Communication				
1.5	Analysis of the existing system	2.5	Competence and awareness				
1.6	Information security policy	2.6	Management of security operations				
1.7	Risk management						
1.8	Statement of Applicability						

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ISO/IEC 27001's Requirements for Internal Audit

ISO/IEC 27001, clauses 9.2.1 and 9.2.2

The organization shall conduct internal audits at planned intervals to provide information on whether the information security management system:

a) conforms to

1) the organization's own requirements for its information security management system;

2) the requirements of this document;

b) is effectively implemented and maintained.

The organization shall plan, establish, implement and maintain an audit program(s), including the frequency, methods, responsibilities, planning requirements and reporting.

When establishing the internal audit program(s), the organization shall consider the importance of the processes concerned and the results of previous audits.

The organization shall:

a) define the audit criteria and scope for each audit;

b) select auditors and conduct audits that ensure objectivity and the impartiality of the audit process;

c) ensure that the results of the audits are reported to relevant management;

Documented information shall be available as evidence of the implementation of the audit program(s) and the audit results.

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ISO/IEC 27003, clause 9.2 Internal audit

Auditors also evaluate whether the ISMS is effectively implemented and maintained. An audit program describes the overall framework for a set of audits, planned for specific time frames and directed towards specific purposes. This is different from an audit plan, which describes the activities and arrangements for a specific audit. Audit criteria are a set of policies, procedures or requirements used as a reference against which audit evidence is compared, i.e. the audit criteria describe what the auditor expects to be in place.

If the outcome of the audit includes nonconformities, the auditee should prepare an action plan for each nonconformity to be agreed with the audit team leader. A follow-up action plan typically includes:

i.*description of the detected nonconformity;*

j.*description of the cause(s) of nonconformity;*

k.*description of short term correction and longer term corrective action to eliminate a detected nonconformity within a defined timeframe; and*

l.*the persons responsible for implementing the plan.*

Audit reports, with audit results, should be distributed to top management.

Results of the previous audits should be reviewed and the audit program adjusted to better manage areas experiencing higher risks due to nonconformity.

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What Is an Audit?

ISO 19011, clause 3.1

Systematic, independent and documented process for obtaining objective evidence and evaluating it objectively to determine the extent to which the audit criteria are fulfilled

Simply put, auditing is asking the auditee what they do and how they do it, in order to check whether their practices are in compliance with organizational policies, procedures, and processes as well as national and international standard requirements.

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An audit is an assessment based on evidence and facts. This assessment points out the strengths and weaknesses of the management system. Audit results are communicated to the organization's top management, who then undertake the required and appropriate measures.

- A financial audit determines whether an organization's accounting practices comply with legal requirements and recognized principles.
- An administrative audit determines the effectiveness of the overall administrative practices of an organization.
- An ISMS audit determines the effectiveness of information security processes and controls in an organization.

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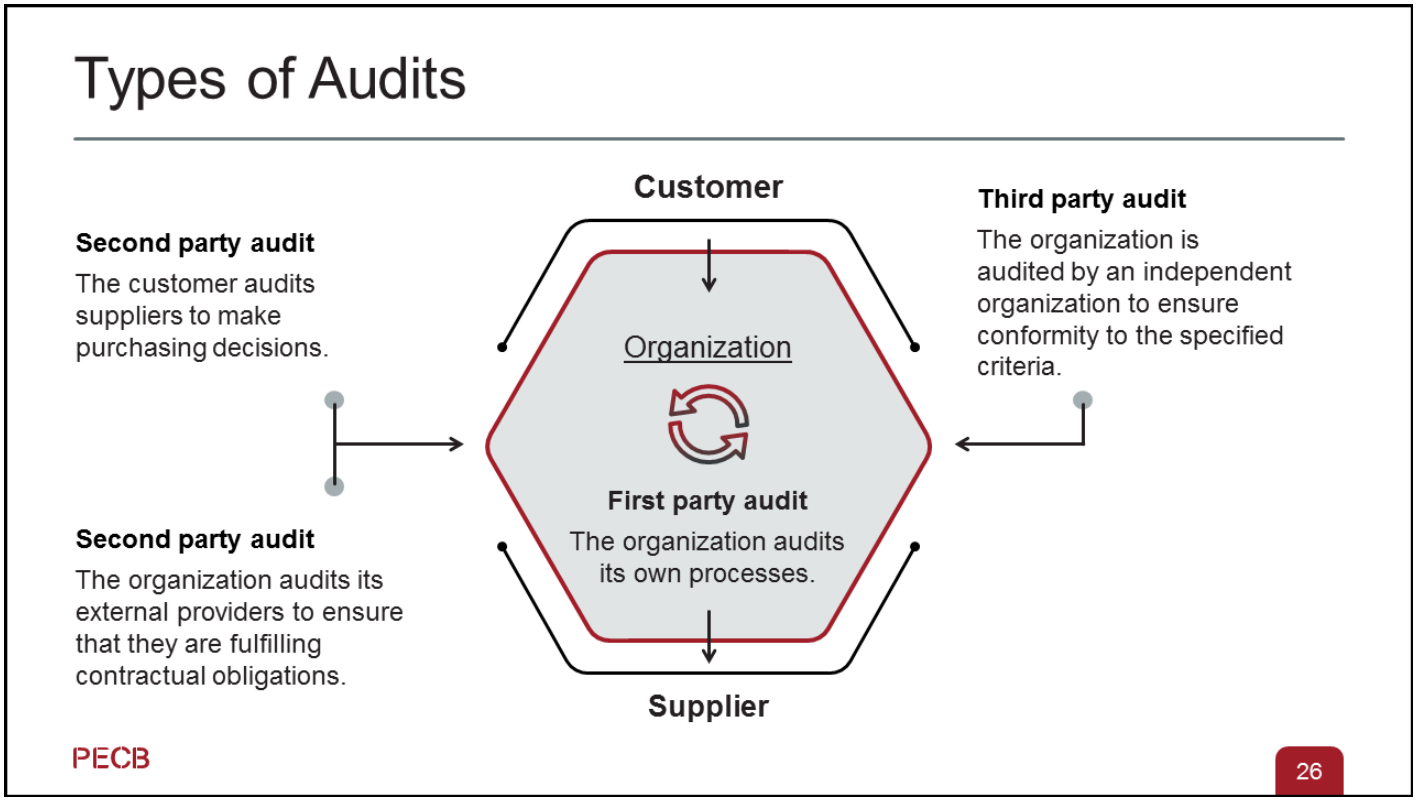

First party audits are best known as internal audits. Internal audit is an independent and objective activity that gives an organization assurance on the level of control over operations, gives recommendations to improve operations, and contributes to creating added value. Internal audits are conducted by the organization itself for the purpose of management reviews and other internal needs.

Second and third party audits are both known as external audits. The main difference between the two is in certification. On the one hand, second party audits are conducted by parties that have an interest in the organizations that is being audited (e.g., customers) and is not intended for certification purposes. On the other hand, third party audits are conducted by external and independent audit organizations (e.g., certification bodies, governmental agencies) and the aim is to obtain certification.

Note: Third party audits are performed by auditors who are external to and independent of the organization being audited.

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Differences between Internal and External Audits

Internal audit

1. Independent of the audited activities (not of the organization)
2. Considers the effectiveness and efficiency of the ISMS
3. Advisory role within the organization for the improvement of the ISMS
4. May be conducted on an ongoing basis

External audit

1. Independent of the organization being audited and its activities
2. Considers only the effectiveness of the ISMS
3. No advisory role within the organization
4. Always conducted in a planned and timely manner

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Internal audits are independent, objective, and advisory activities performed with the aim of improving an organization's functions. In addition, internal audits contribute to the objectives of the organization by providing a systematic and structured methodology to evaluate and improve the effectiveness of the risk management process, its control, and decision-making.

Internal audits aim to safeguard the organization's assets, promote operational efficiency, check the accuracy and reliability of information, and encourage adherence to policies and procedures.

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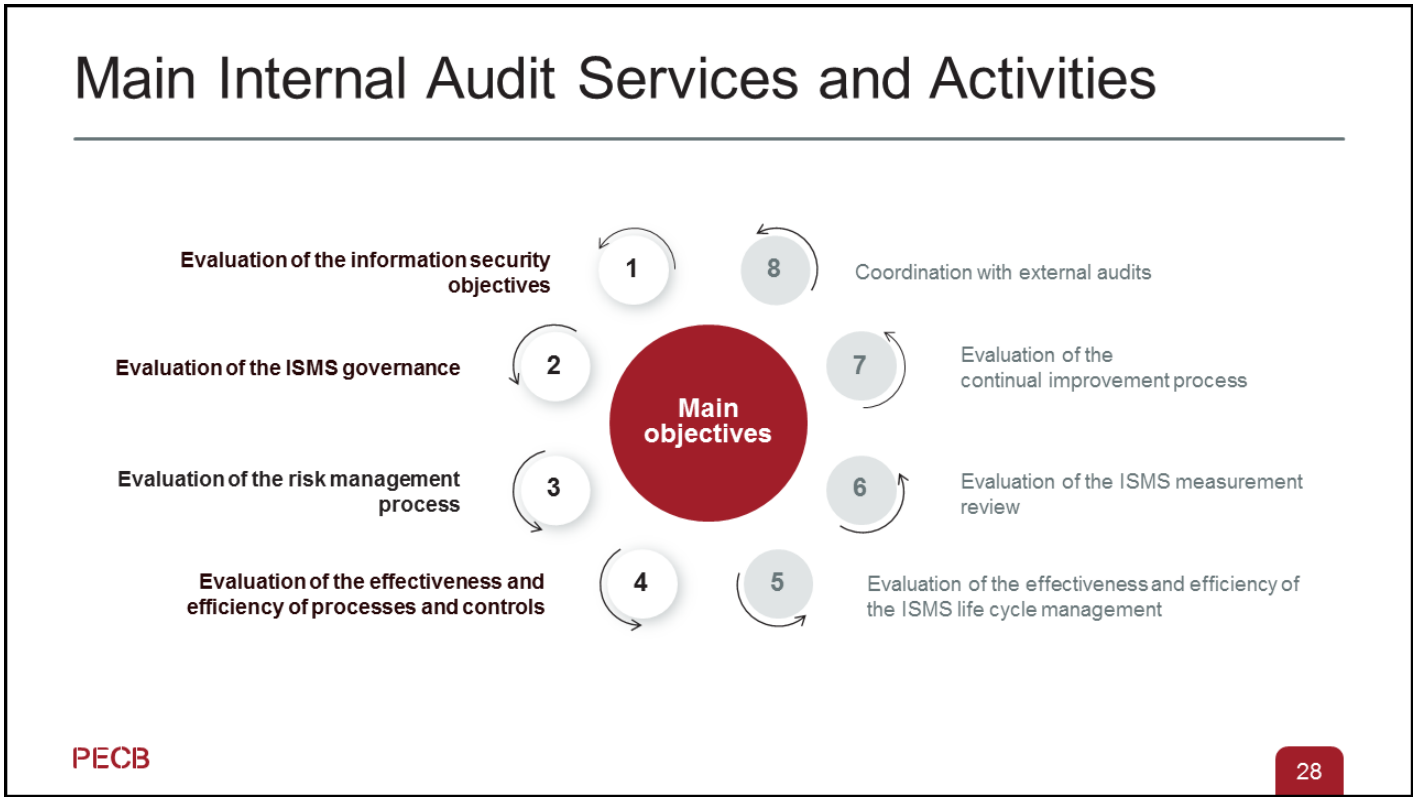

The objectives of the internal audit should be reviewed and approved by the organization's top management. In the context of a management system certification, the objectives of the internal audit should at least cover the evaluation of activities related to the management system. However, the internal audit may cover several other audit activities such as financial, administrative, quality assurance, etc. The objectives of the internal audit are defined based on the size of the organization, its sector of activity, and its mission.

The main activities of the internal audit are:

1. **Evaluation of the information security objectives:** The internal auditor should evaluate whether the organization's information security objectives are well aligned with its business plan and strategic direction.
2. **Evaluation of the ISMS governance:** The internal auditor should validate if the organization's management supports activities related to the management system and whether the roles and responsibilities of interested parties are clearly defined.
3. **Evaluation of the risk management process:** The internal auditor should evaluate whether the organization has implemented and maintains an ongoing risk management with regard to the ISMS. Unlike an external auditor, an internal auditor may participate as an interested party in identifying and assessing the risks faced by the organization.
4. **Evaluation of the effectiveness and efficiency of processes and controls:** The internal auditor should evaluate the adequacy, effectiveness, and efficiency of information security processes and controls in operation to determine whether they are in line with the normative, legal, regulatory, and contractual requirements, as well as with the internal policies of the organization.

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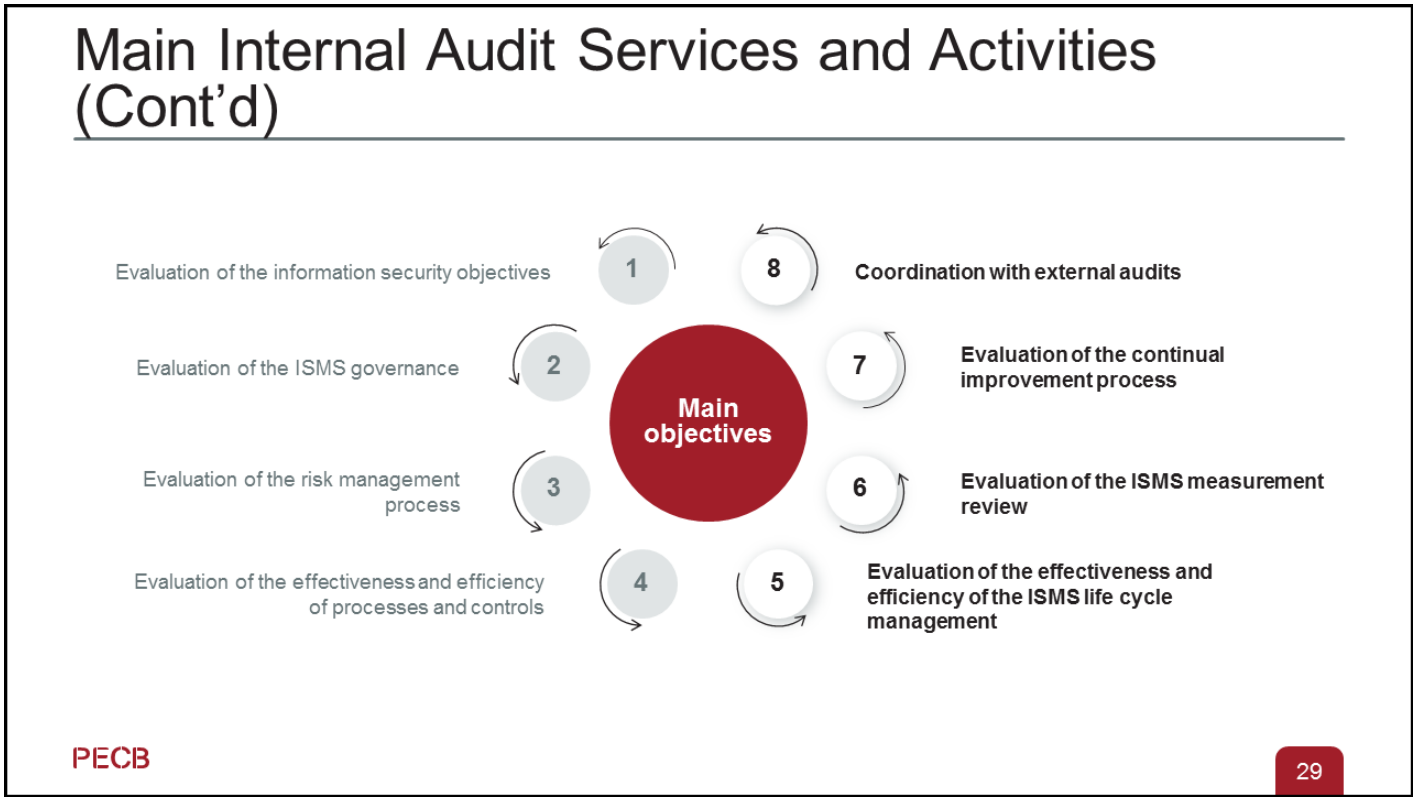

5.Evaluation of the effectiveness and efficiency of the ISMS life cycle management: The internal auditor should evaluate the effectiveness and efficiency of the life cycle management processes and safety measures related to information security matters including the planning, preparation, implementation, operation, monitoring, review, update, and improvement of the management system.

6.Evaluation of the ISMS measurement review: The internal auditor should make sure that the organization periodically conducts a measurement review of the management system to validate whether the objectives of the organization are met.

7.Evaluation of the continual improvement process: The internal auditor should make sure that the organization is implementing corrective and preventive actions to address its nonconformities and to improve the effectiveness and efficiency of the ISMS.

8.Coordination with external audits: The internal auditor should check whether the internal audit and the external audit activities are well coordinated. The aim of the internal audit is to ensure that the organization performs adequate monitoring of external audit reports and action plans that they have established and approved.

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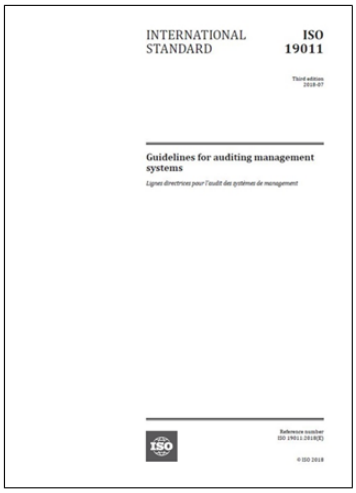
ISO 19011

- ISO 19011 provides guidance on managing an audit program, planning and conducting management system audits, as well as on the competence of auditors.
- This standard explains the fundamental principles of auditing.
- It focuses on first and second party audits.
- Organizations cannot obtain certification against this standard.


INTERNATIONAL STANDARD

ISO 19011

Guidelines for auditing management systems

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The guidelines of this standard are intended to be flexible, so they can be easily adapted to the size, nature, and complexity of the organization to be audited. The responsibility of properly applying the guidelines falls on each auditor.

ISO 19011:2018 has the following structure:

1. *Scope*
2. *Normative references*
3. *Terms and definitions*
4. *Principles of auditing*
5. *Managing an audit program*
6. *Conducting an audit*
7. *Competence and evaluation of auditors*

Annex A (informative) Additional guidance for auditors planning and conducting audits

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3.2 Internal Audit

List of activities

- 3.2.1 Plan an internal audit program
- 3.2.2 Appoint a competent person
- 3.2.3 Ensure independence, objectivity, and impartiality
- 3.2.4 Plan audit activities
- 3.2.5 Allocate and manage resources
- 3.2.6 Perform audit activities
- 3.2.7 Document nonconformities
- 3.2.8 Follow up on nonconformities

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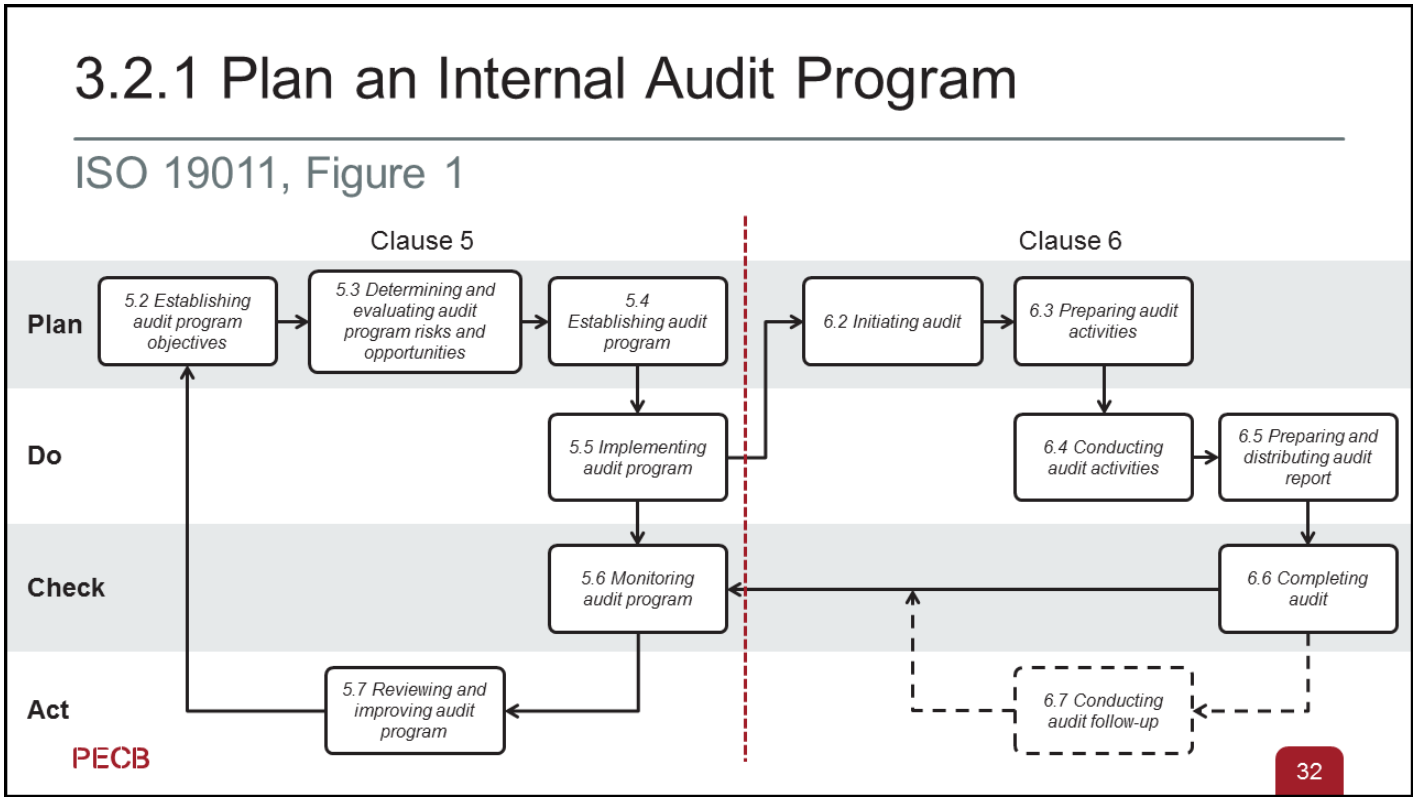

ISO 19011, clause 3.4 Audit program

Arrangements for a set of one or more audits planned for a specific time frame and directed towards a specific purpose

The audit program should follow the steps of the PDCA (Plan-Do-Check-Act) model. The audit program can include more than one audit depending on the size and complexity of the organization. Joint or combined audits can be conducted as well.

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3.2.2 Appoint a Competent Person

Roles and responsibilities of an internal auditor

- Develop an internal audit program (procedures, work papers, etc.)
- Plan the audit activities
- Manage the resources
- Develop the criteria for the internal audit
- Draft audit reports
- Monitor, review, and improve the internal audit program
- Follow up on nonconformities and recommendations from previous audits
- Prepare for external audits

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ISO 19011, clause 5.4.1 Roles and responsibilities of the individual(s) managing the audit program

The individual(s) managing the audit program should:

a. *establish the extent of the audit program according to the relevant objectives and any known constraints;*
b. *determine the external and internal issues, and risks and opportunities that can affect the audit program, and implement actions to address them, integrating these actions in all relevant auditing activities, as appropriate;*
c. *ensuring the selection of audit teams and the overall competence for the auditing activities by assigning roles, responsibilities and authorities, and supporting leadership, as appropriate;*
d. *establish all relevant processes including processes for:*
    - *the coordination and scheduling of all audits within the audit program;*
    - *the establishment of audit objectives, scope(s) and criteria of the audits, determining audit methods and selecting the audit team;*
    - *evaluating auditors;*
    - *the establishment of external and internal communication processes, as appropriate;*
    - *the resolutions of disputes and handling of complaints;*
    - *audit follow-up if applicable;*
    - *reporting to the audit client and relevant interested parties, as appropriate.*
e. *determine and ensure provision of all necessary resources;*
f. *ensure that appropriate documented information is prepared and maintained, including audit program records;*
g. *monitor, review and improve the audit program;*
h. *communicate the audit program to the audit client and, as appropriate, relevant interested parties.*

The individual(s) managing the audit program should request its approval by the audit client.

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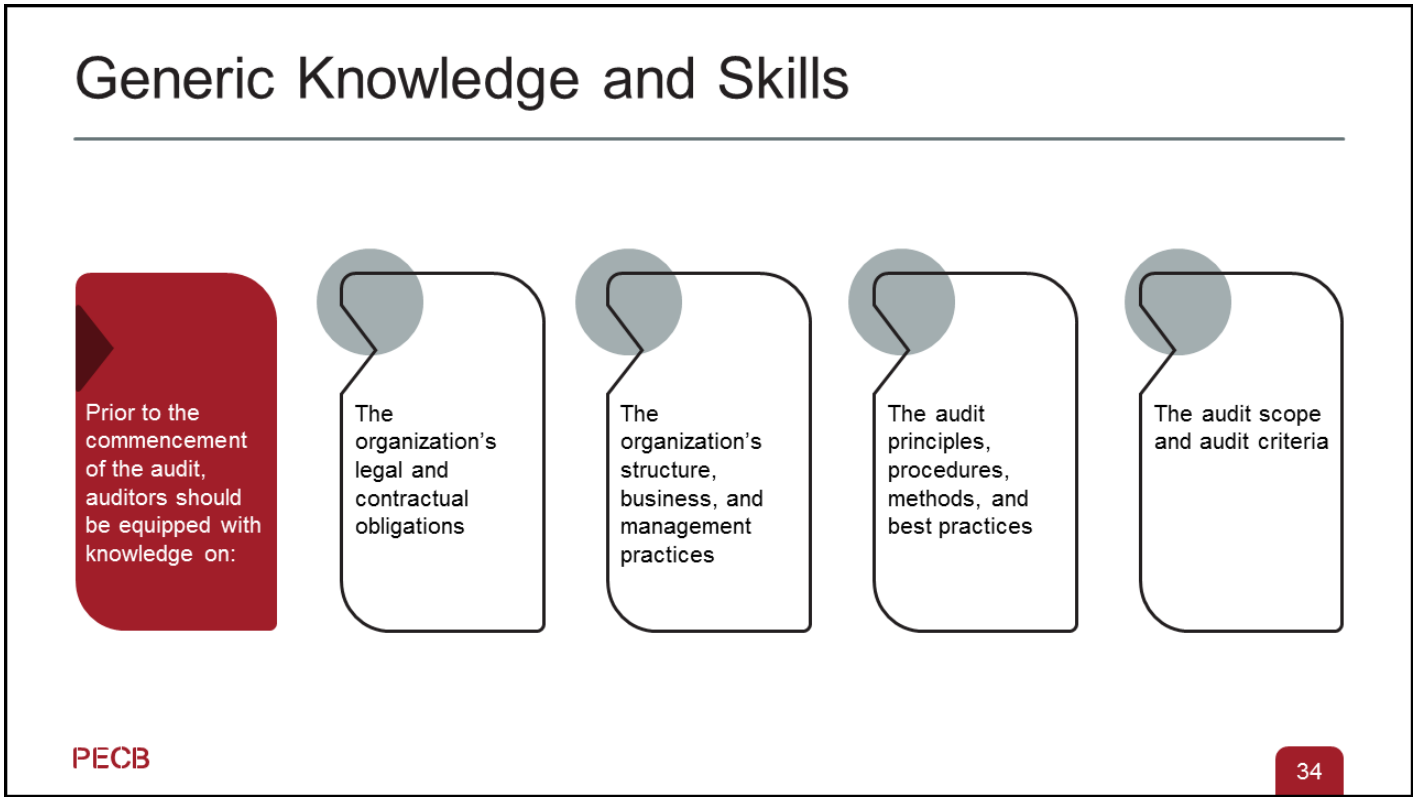

ISO 19011, clause 7.2.3.2 Generic knowledge and skills of management system auditors

Auditors should have knowledge and skills in the areas outlined below.

a.*Audit principles, processes and methods: knowledge and skills in this area enable the auditor to ensure audits are performed in a consistent and systematic manner.*

An auditor should be able to:

- *understand the types of risks and opportunities associated with auditing and the principles of the risk-based approach to auditing;*
- *plan and organize the work effectively;*
- *perform the audit within the agreed time schedule;*
- *prioritize and focus on matters of significance;*
- *communicate effectively, orally and in writing (either personally, or through the use of interpreters);*
- *collect information through effective interviewing, listening, observing and reviewing documented information, including records and data;*
- *understand the appropriateness and consequences of using sampling techniques for auditing;*
- *understand and consider technical experts' opinions;*
- *audit a process from start to finish, including the interrelations with other processes and different functions, where appropriate;*
- *verify the relevance and accuracy of collected information;*
- *confirm the sufficiency and appropriateness of audit evidence to support audit findings and conclusions;*
- *assess those factors that may affect the reliability of the audit findings and conclusions;*
- *document audit activities and audit findings, and prepare reports;*
- *maintain the confidentiality and security of information.*

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Slide Notes Extension

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ISO 19011, clause 7.2.3.2 Generic knowledge and skills of management system auditors (cont'd)

b.*Management system standards and other references: knowledge and skills in this area enable the auditor to understand the audit scope and apply audit criteria, and should cover the following:*

- *management system standards or other normative or guidance/supporting documents used to establish audit criteria or methods;*
- *the application of management system standards by the auditee and other organizations;*
- *relationships and interactions between the management system(s) processes;*
- *understanding the importance and priority of multiple standards or references;*
- *application of standards or references to different audit situations.*

c.*The organization and its context: knowledge and skills in this area enable the auditor to understand the auditee's structure, purpose and management practices and should cover the following:*

- *needs and expectations of relevant interested parties that impact the management system;*
- *type of organization, governance, size, structure, functions and relationships;*
- *general business and management concepts, processes and related terminology, including planning, budgeting and management of individuals;*
- *cultural and social aspects of the auditee.*

d.*Applicable statutory and regulatory requirements and other requirements: knowledge and skills in this area enable the auditor to be aware of, and work within, the organization's requirements. Knowledge and skills specific to the jurisdiction or to the auditee's activities, processes, products and services should cover the following:*

- *statutory and regulatory requirements and their governing agencies;*
- *basic legal terminology;*
- *contracting and liability.*

NOTE Awareness of statutory and regulatory requirements does not imply legal expertise and a management system audit should not be treated as a legal compliance audit.

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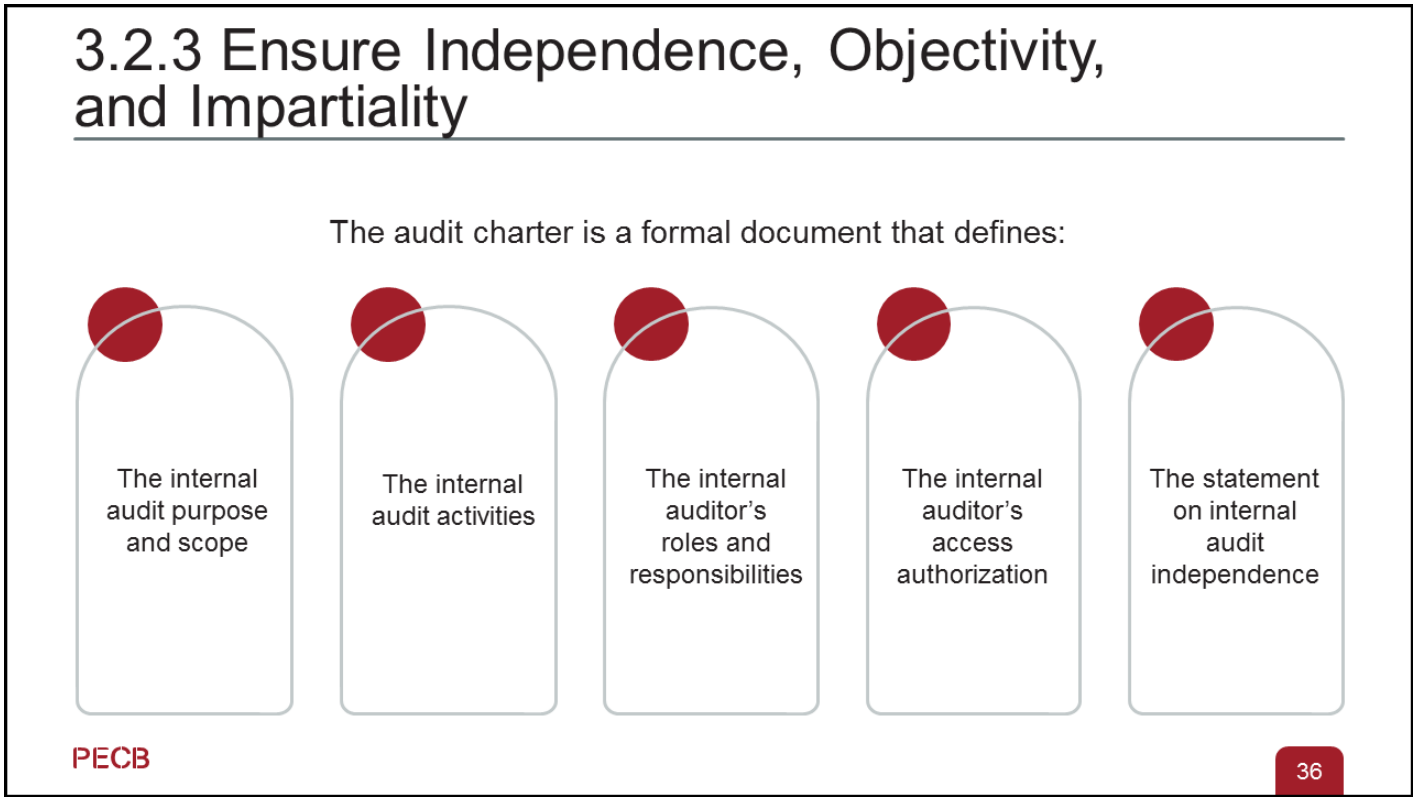

While internal auditors are not independent of the organizations that employ them, independence, objectivity, and impartiality are the cornerstone of the internal audit profession. This independence enables unrestricted evaluation of management activities and personnel and allows internal auditors to perform their role successfully.

To ensure the objectivity and impartiality of the internal audit function, auditors should not undertake operational roles related to the management system. If a person has assumed such a role, a reasonable period of time (usually one year) should pass before the person can occupy the position of the internal auditor. A person may undertake operational roles and conduct an audit only if the two spheres of activities involved are not related. In this case, there have to be well-documented job descriptions to avoid potential conflicts of interest and violations of the principle of independence.

The internal audit charter is a way of describing how the organization will benefit from the internal audit especially in achieving its objectives. The top management should approve the internal audit charter.

Important note: Smaller organization may find it better to outsource the internal audit function to a third party. It is indeed easier to demonstrate the independence and impartiality of a person who has no connection with the implementation and operations of the ISMS.

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Access and Independence

Principles

1

Access to resources and collaboration

- Internal auditors should have unrestricted access to executives, employees, information, explanations, and documented information necessary for the proper conduct of the audit.
- This demand for access must be documented (usually in the audit charter).

2

Independence

- Internal auditors must be independent of the processes being audited; this is ensured if internal auditors report directly to the organization's audit board rather than to top management.
- This demand for independence should be reflected in the organizational chart.

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In order to ensure that the mission of the internal auditor is successfully completed, the organization must demonstrate their availability and collaboration.

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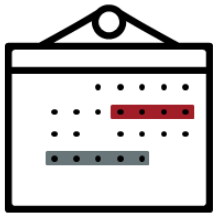
3.2.4 Plan Audit Activities

The main activities that an internal auditor must plan beforehand include:

- Reviewing the application of internal policies and procedures
- Reviewing the effectiveness and efficiency of operations
- Reviewing the use of resources in the organization
- Evaluating the degree of conformity to laws, regulations, and standard requirements
- Reviewing the information security risk assessment process
- Reviewing the organization's documented information to verify the allocation of the roles and responsibilities for information security



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3.2.5 Allocate and Manage Resources

The success of an audit greatly depends on the effective allocation and management of resources. Some major resources that are needed for the conduct of an audit are as follows:

Financial resources | Competent personnel | Tools | Audit policies and procedures | Logistics

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ISO 19011, clause 5.4.4 Determining audit program resources

When determining resources for the audit program, the individual(s) managing the audit program should consider:

a. *the financial and time resources necessary to develop, implement, manage and improve audit activities;*
b. *audit methods;*
c. *the individual and overall availability of auditors and technical experts having competence appropriate to the particular audit program objectives;*
d. *the extent of the audit program and audit program risks and opportunities;*
e. *travel time and cost, accommodation and other auditing needs;*
f. *the impact of different time zones;*
g. *the availability of information and communication technologies (e.g. technical resources required to set up a remote audit using technologies that support remote collaboration);*
h. *the availability of any tools, technology and equipment required;*
i. *the availability of necessary documented information, as determined during the establishment of the audit program;*
j. *requirements related to the facility, including any security clearances and equipment (e.g. background checks, personal protective equipment, ability to wear clean room attire).*

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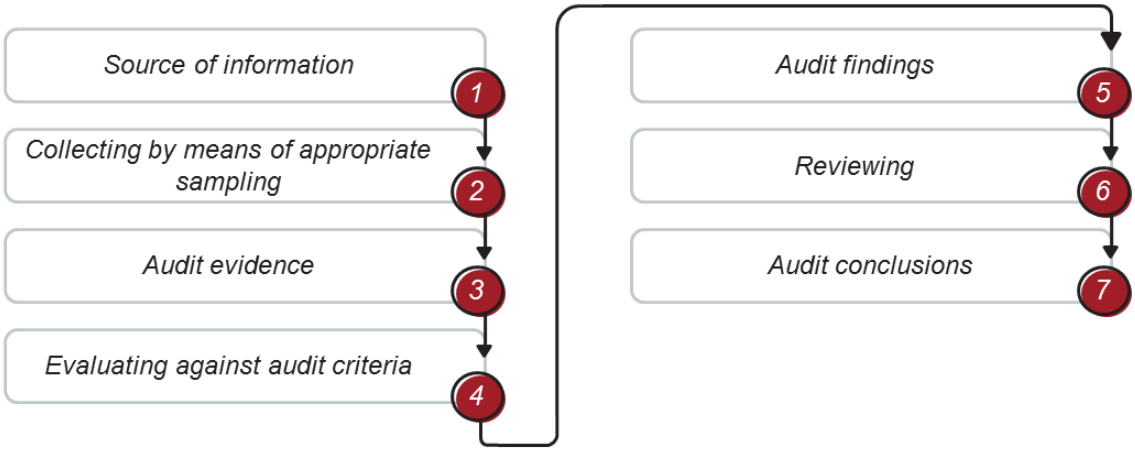
3.2.6 Perform Audit Activities

ISO 19011, Figure 2

Overview of a typical process of collecting and verifying information



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ISO 19011, clause 3.9 Audit evidence

Records, statements of fact or other information, which are relevant to the audit criteria and verifiable

ISO 19011, clause 3.10 Audit findings

Results of the evaluation of the collected audit evidence against audit criteria

Note 1 to entry: Audit findings indicate conformity or nonconformity.

Note 2 to entry: Audit findings can lead to the identification of risks, opportunities for improvement or recording good practices.

Note 3 to entry: In English if the audit criteria are selected from statutory requirements or regulatory requirements, the audit finding is termed compliance or non-compliance.

ISO 19011, clause 3.11 Audit conclusion

Outcome of an audit, after consideration of the audit objectives and all audit findings

To ensure the relevance of an audit procedure, the auditor must collect evidence from different sources of information and evaluate them objectively. The evidence-collection process can be carried out by using different audit procedures and methods, including sampling, when and if required.

After evaluating the audit evidence against the audit criteria, the auditor drafts the audit findings. Finally, following the analysis of all the audit findings and reviewing them, the auditor issues the audit conclusion(s).

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Nonconformity

Definition

- According to ISO 9000, a nonconformity is defined as the “non-fulfillment of a requirement.”
- There are two types of nonconformities:
  - Minor nonconformity
  - Major nonconformity

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Requirements can derive from several sources: national or international standards, internal rules and policies of the organization, laws and regulations of the country in which the organization operates, contracts signed with clients or partners.

ISO 9000, clause 3.6.11 Conformity

Fulfillment of a requirement

Common examples of nonconformities include:

- The documentation is not complete.
- The control is not implemented or is ineffective.

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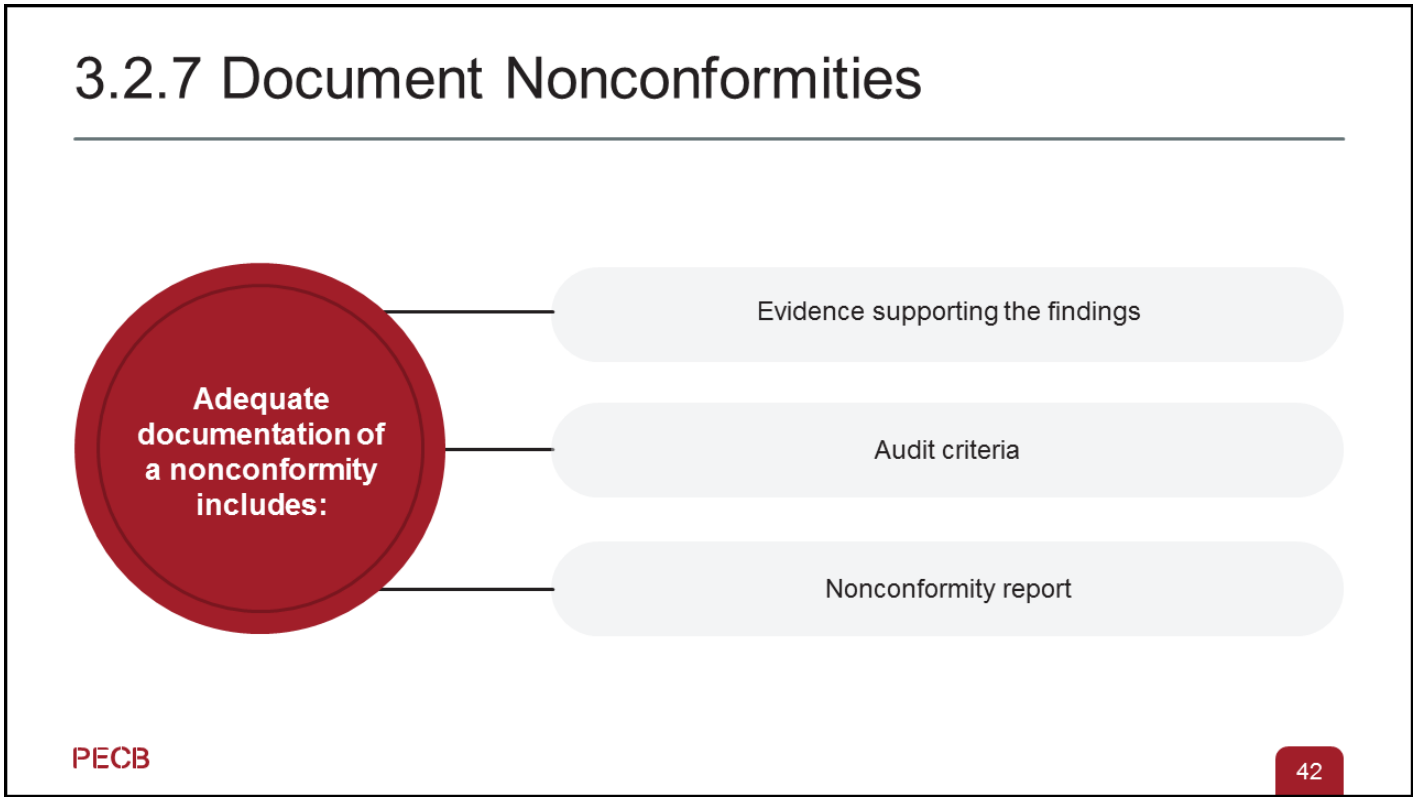

Once a nonconformity has been identified, the auditor must document it. The recording of the nonconformity can be as simple as a description of the observation and the reference to the audit criteria.

A nonconformity report should be explicit, unambiguous, linguistically correct, and as concise as possible.

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Nonconformity Report

Example

NONCONFORMITY REPORT		
Nonconformity number: 3	**Client:** Thalia Technologies	**File number:** 34527
Process: Organizational roles, responsibilities, and authorities	**Clause number:** 5.3	**Site:** Montreal
Audit criteria: *Top management shall ensure that the responsibilities and authorities for roles relevant to information security are assigned and communicated.*		
Description of the observed nonconformity: In a sample of 30 interviewed employees, it was found that only 5 of them are aware of their responsibilities regarding information security.		
Auditor: J. Rogers	**Acknowledgment by auditee representative:** Nonconformity presented to Mr. R. Smith and confirmed on March 21, 2022	**Nonconformity**
		Major
Date: March 20, 2022		Minor

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3.2.8 Follow up on Nonconformities

- An internal auditor should follow up on action plans submitted in response to nonconformities.
- The person in charge of the ISMS must inform the internal auditor of the progress of corrections and corrective actions taken.
- The internal auditor should review the corrections, the identified root causes, and the corrective actions, and verify the effectiveness of all corrections and corrective actions.
- Not all corrections and corrective actions have to be implemented immediately.

Note: The auditor, based on experience and knowledge, should exercise good judgment and assess whether the action plans are appropriate and whether they can address the root causes of the nonconformities.

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The auditor must always remember that it is highly unlikely for an organization to correct all nonconformities simultaneously. Every correction or corrective action requires the use of resources and time for implementation. Action plans can be arranged in order of priority by the management, especially for those that require investment. Therefore, the auditor must ensure that the objectives regarding the correction of nonconformities are realistic.

The auditor should request or receive a periodic update from the organization to assess the progress that has been made regarding the correction of nonconformities. The monitoring of action plans is particularly important with respect to the high risk problems and corrective actions with long lead times.

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Section Summary:

- Internal audits help organizations determine if their ISMS is effective efficient, and if it fulfills the requirements of ISO/IEC 27001.
- Internal audit results are used as input for continual improvement.
- The internal auditor is responsible for developing an internal audit program, planning the audit activities, drafting audit reports, managing the resources, establishing the criteria for the internal audit, monitoring, reviewing, and improving the internal audit program, following up on nonconformities, and preparing the organization for external audits.
- Major resources needed to conduct an internal audit include: financial resources, competent personnel, tools, audit policies and procedures, and logistics.
- The auditor should follow up on nonconformities by reviewing the corrections and corrective actions, and by verifying their effectiveness.

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Questions?

Quiz 23

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Note: To complete Quiz 23, please go to the Quizzes Sheet.

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Section 23

Management review

- Preparing for the management review
- Conducting the management review
- Determining the management review outputs
- Following up on the management review

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This section will help the participant gain knowledge on preparing and conducting a management review. In addition, the participant will be able to understand the process of closing the review and the follow-up activities of the management review.

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Management Review

1. Define and establish		2. Implement and operate		3. Monitor and review		4. Maintain and improve	
1.1	Understanding the organization and its context	2.1	Selection and design of controls	3.1	Monitoring, measurement, analysis, and evaluation	4.1	Treatment of nonconformities
1.2	ISMS scope	2.2	Implementation of controls	3.2	Internal audit	4.2	Continual improvement
1.3	Leadership and project approval	2.3	Management of documented information	3.3	**Management review**		
1.4	Organizational structure	2.4	Communication				
1.5	Analysis of the existing system	2.5	Competence and awareness				
1.6	Information security policy	2.6	Management of security operations				
1.7	Risk management						
1.8	Statement of Applicability						

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ISO/IEC 27001's Requirements for Management Reviews

ISO/IEC 27001, clauses 9.3.1 and 9.3.2

Top management shall review the organization's information security management system at planned intervals to ensure its continuing suitability, adequacy and effectiveness.

The management review shall include consideration of:

a) *the status of actions from previous management reviews;*
b) *changes in external and internal issues that are relevant to the information security management system;*
c) *changes in needs and expectations of interested parties that are relevant to the information security management system;*
d) *feedback on the information security performance, including trends in:*
   1) *nonconformities and corrective actions;*
   2) *monitoring and measurement results;*
   3) *audit results;*
   4) *fulfilment of information security objectives;*
e) *feedback from interested parties;*
f) *results of risk assessment and status of risk treatment plan;*
g) *opportunities for continual improvement.*

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ISO/IEC 27001, clause 9.3.3 Management review results

The results of the management review shall include decisions related to continual improvement opportunities and any needs for changes to the information security management system.

Documented information shall be available as evidence of the results of management reviews.

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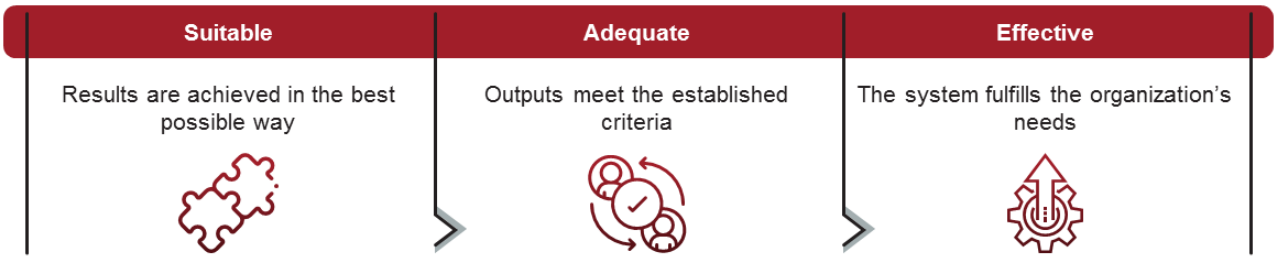
ISO/IEC 27001's Requirements for Management Reviews

ISO/IEC 27001, clause 9.3

Top management shall review the organization's information security management system at planned intervals to ensure its continuing suitability, adequacy and effectiveness.



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ISO/IEC27001, clause9.3 Management review (cont'd)

The management review shall include consideration of:

a. *the status of actions from previous management reviews;*
b. *changes in external and internal issues that are relevant to the information security management system;*
c. *feedback on the information security performance, including trends in:*
    1. *nonconformities and corrective actions;*
    2. *monitoring and measurement results;*
    3. *audit results; and*
    4. *fulfilment of information security objectives;*
d. *feedback from interested parties;*
e. *results of risk assessment and status of risk treatment plan; and*
f. *opportunities for continual improvement.*

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3.3 Management Review

List of activities

- 3.3.1 Prepare for the management review
- 3.3.2 Conduct the management review
- 3.3.3 Determine the management review outputs
- 3.3.4 Follow up on the management review

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3.3.1 Prepare for the Management Review

- Management reviews must be conducted at planned intervals (e.g., monthly, quarterly semi-annually, or annually). Although ISO/IEC 27001 does not provide specific requirements regarding the frequency of management review meetings, it is common practice to conduct management review each quarter.
- Management reviews can be included in a regular management meeting and discussed as a topic on the agenda.
- It is good practice to send all related documentation to the attendees (e.g., audit report, results of reviews, action plans) before the management review meeting.

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3.3.2 Conduct the Management Review

ISO 27003, clause 9.3

The agenda of the management review should address the following topics:

a) *status of actions from previous management reviews;*
b) *changes in external and internal issues that are relevant to the ISMS;*
c) *feedback on the information security performance, including trends, in:*
   1) *nonconformities and corrective actions;*
   2) *monitoring and measurement results;*
   3) *audit results; and*
   4) *fulfilment of information security objectives.*
d) *feedback from interested parties, including suggestions for improvement, requests for change and complaints;*
e) *results of information security risk assessment(s) and status of information security risk treatment plan; and*
f) *opportunities for continual improvement, including efficiency improvements of both the ISMS and information security controls.*

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The input needed for each of the topics mentioned in the slide must be prepared in advance. The aim of the review will be to verify whether the defined objectives by management are being accomplished and whether the ISMS is meets the requirements of the standard. For each point, the management review will decide what (if any) actions to take.

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3.3.3 Determine the Management Review Outputs

ISO/IEC 27001, clause 9.3

The outputs of the management review shall include decisions related to:

continual improvement activities **and** *any needs for changes to the information security management system.*

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3.3.4 Follow Up on the Management Review

ISO/IEC 27001, clause 9.3

The organization shall retain documented information as evidence of the results of management reviews.

Information from management reviews that should be documented includes:

- The date and location of review
- Contributors and their role to the review
- The criteria against which the ISMS has been reviewed
- Current performance of the ISMS
- Any changes identified in the internal and external environment
- Decisions and actions to be taken
- Responsibilities and timescales for the agreed actions

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Section Summary:

- Management review is conducted by the top management to analyze the suitability, adequacy, and effectiveness of the information security management system. Management reviews must be conducted at planned intervals.
- Management review should include, among others, information on audit results, nonconformities and corrective actions, review of new and ongoing actions, results of monitoring and measurement, risks assessment, and status of risk treatment plan.
- Management review outputs include decisions in regard to continual improvement opportunities and changes to the ISMS.

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Questions?

Quiz 24

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This section aims to help participants gain knowledge on treating nonconformities and understand the importance of corrective and preventive actions. Moreover, participants will learn how to draft an action plan and review the effectiveness of the actions taken.

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Treatment of Nonconformities

1. Define and establish	2. Implement and operate	3. Monitor and review	4. Maintain and improve
1.1 Understanding the organization and its context	2.1 Selection and design of controls	3.1 Monitoring, measurement, analysis, and evaluation	4.1 **Treatment of nonconformities**
1.2 ISMS scope	2.2 Implementation of controls	3.2 Internal audit	4.2 Continual improvement
1.3 Leadership and project approval	2.3 Management of documented information	3.3 Management review	
1.4 Organizational structure	2.4 Communication		
1.5 Analysis of the existing system	2.5 Competence and awareness		
1.6 Information security policy	2.6 Management of security operations		
1.7 Risk management			
1.8 Statement of Applicability			

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ISO/IEC 27001's Requirements for Nonconformities and Corrective Actions

ISO/IEC 27001, clause 10.2

When a nonconformity occurs, the organization shall:

a) react to the nonconformity, and as applicable:
  1) take action to control and correct it;
  2) deal with the consequences;

b) evaluate the need for action to eliminate the causes of nonconformity, in order that it does not recur or occur elsewhere, by:
  1) reviewing the nonconformity;
  2) determining the causes of the nonconformity; and
  3) determining if similar nonconformities exist, or could potentially occur;

c) implement any action needed;

d) review the effectiveness of any corrective action taken; and

e) make changes to the information security management system, if necessary.

Corrective actions shall be appropriate to the effects of the nonconformities encountered.

Documented information shall be available as evidence of:

f) the nature of the nonconformities and any subsequent actions taken,

g) the results of any corrective action.

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ISO/IEC 27003, clause 10.1 Nonconformity and corrective action

Explanation

A nonconformity is a non-fulfilment of a requirement of the ISMS. Requirements are needs or expectations that are stated, implied or obligatory. There are several types of nonconformities such as:

a.*failure to fulfill a requirement (completely or partially) of ISO/IEC 27001 in the ISMS;*

b.*failure to correctly implement or conform to a requirement, rule or control stated by the ISMS; and*

c.*partial or total failure to comply with legal, contractual or agreed customer requirements.*

Nonconformities can be for example:

d.*persons not behaving as expected by procedures and policies;*

e.*suppliers not providing agreed products or services;*

f.*projects not delivering expected outcomes; and*

g.*controls not operating according to design.*

Nonconformities can be recognized by:

h.*deficiencies of activities performed in the scope of the management system;*

i.*ineffective controls that are not remediated appropriately;*

j.*analysis of information security incidents, showing the non-fulfilment of a requirement of the ISMS;*

k.*complaints from customers;*

l.*alerts from users or suppliers;*

m.*monitoring and measurement results not meeting acceptance criteria; and*

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n.*objectives not achieved.*

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Definitions

ISO 9000, clauses 3.3.2, 3.12.3, 3.12.2, and 3.12.1

Continual improvement	***Correction***	***Corrective action***	***Preventive action***
Recurring activity to enhance performance	*Action to eliminate a detected nonconformity*	*Action to eliminate the cause of a nonconformity and to prevent recurrence*	*Action to eliminate the cause of a potential nonconformity or other potential undesirable situation*

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Notes on terminology:

1. By definition, information security improvement is the part of information security management focused on increasing the ability to fulfill information security requirements. The requirements can be related to any aspect, including effectiveness, efficiency, or traceability.
2. The process of establishing objectives and finding opportunities for improvement is a continual process that uses audit findings and audit conclusions, analysis of data, management reviews, or other means. It generally leads to corrective action or preventive action.
3. Preventive action is taken to prevent occurrence, whereas corrective action is taken to prevent recurrence.
4. A correction can be made in conjunction with a corrective action.

ISO 9000, clause 3.7.11 Effectiveness

Extent to which planned activities are realized and planned results are achieved

ISO 9000, clause 3.7.10 Efficiency

Relationship between the result achieved and the resources used

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4.1 Treatment of Nonconformities

List of activities

- **4.1.1** Define a process to resolve problems and nonconformities
- **4.1.2** Determine the corrective actions
- **4.1.3** Determine the preventive actions
- **4.1.4** Draft an action plan

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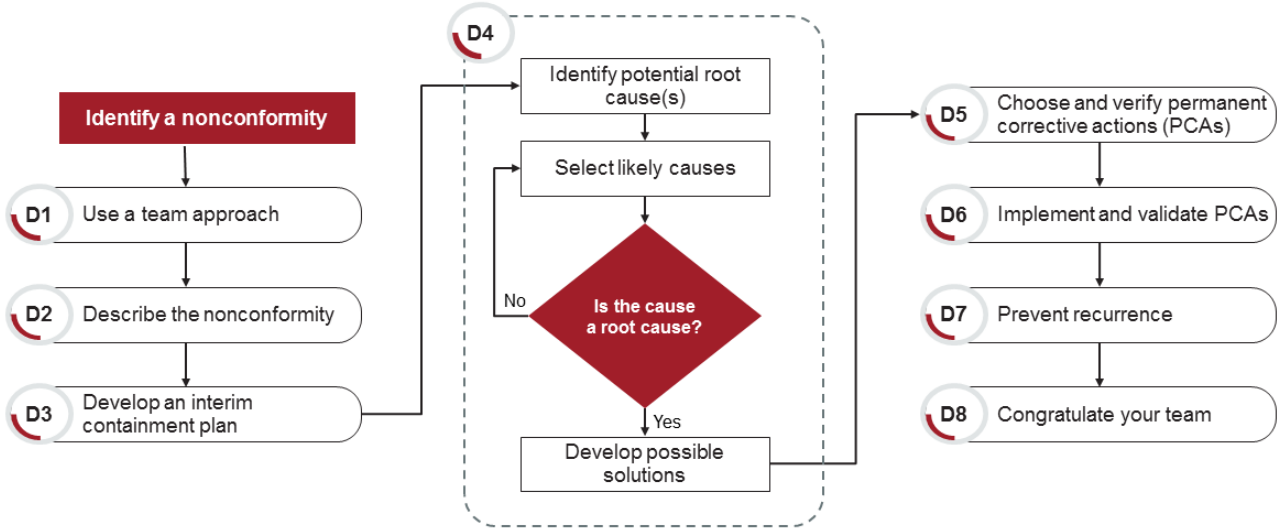
4.1.1 Define a Process to Resolve Problems and Nonconformities

Example of the eight disciplines problem-solving method:



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The Eight Disciplines Problem Solving (8Ds) is a method used to approach and resolve problems and nonconformities. Its purpose is to identify, correct, and eliminate recurring problems. Moreover, it is useful to improve processes and controls. It establishes a permanent corrective action based on a statistical analysis of the problem and focuses on the origin of the problem by determining its root causes.

The steps include:

- **D1 — Use a team approach:** establish a team consisting of professionals with thorough knowledge of processes and controls
- **D2 — Describe the nonconformity:** define the problem by breaking it down into measurable items (The 5W2H [who, what, where, when, why, how and how many] can be used as a tool in this step.)
- **D3 — Develop an interim containment plan:** determine and implement the respective containment actions in order to confine the problem and thus prevent it from affecting the interested parties
- **D4 — Identify potential root cause(s):** identify and analyze all possible reasons for the occurrence of the problem in the first place; seek an explanation as to why the problem has not been detected at the time it occurred (The root causes must be properly verified and, if required, proven, and not simply determined by a brainstorming session.)
- **D5 — Choose and verify permanent corrective actions (PCAs):** make sure, quantitatively, that the chosen corrective action will resolve the problem by means of pre-production programs.
- **D6 — Implement and validate PCAs:** determine and implement the best corrective actions
- **D7 — Prevent recurrence:** in order to avoid the repetition of the same problems or similar ones, take actions by modifying the management systems, operation systems, practices, and procedures
- **D8 — Congratulate your team:** acknowledge the collective efforts of the team members and officially thank them for their work

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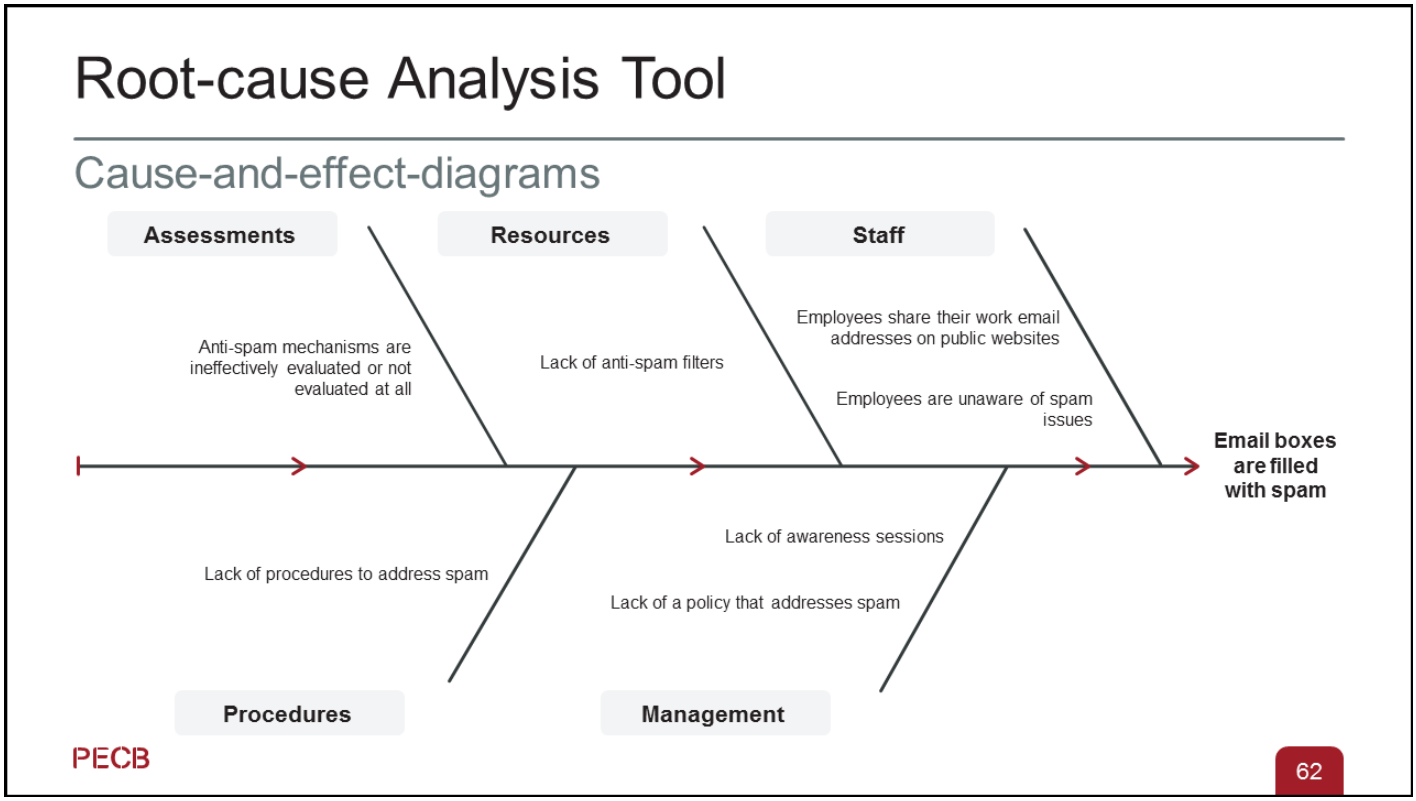

Root cause analysis is a method used to resolve nonconformities by identifying the root causes of the problem. This practice is based on the theory that problems are better resolved when the root causes are corrected or eliminated, as opposed to simply treating the immediate and obvious symptoms.

By addressing the root causes, the possibility of the nonconformity to recur is reduced. To be effective, root cause analysis must be performed systematically and conclusions must be supported by evidence. For each problem analyzed, there is generally more than one root cause.

A cause-and-effect diagram, also known as a fishbone diagram, is a root-cause analysis tool that maps the causes and the effects visually. It sorts possible causes of a problem into different smaller categories to get closer to the root cause of the problem. It is vital to choose the categories wisely in order to identify the root cause of the problem and find the appropriate solution.

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Asking the Right Questions

The questions needed for the analysis of any problem

Current situation	Questioning	Solution tracking	Remaining option(s)
What has been done?	Why is this necessary?	What else could we do?	What will be done?
How is it done?	Why is it done this way?	How to do it differently?	How will it be done?
Who did it?	Why this person?	Who else could do it?	Who will do it?
Where is it done?	Why is it done at this place?	Where else could we do it?	Where will it be done?
When is it done?	Why is it done at that moment?	Could we do it another time?	When will it be done?

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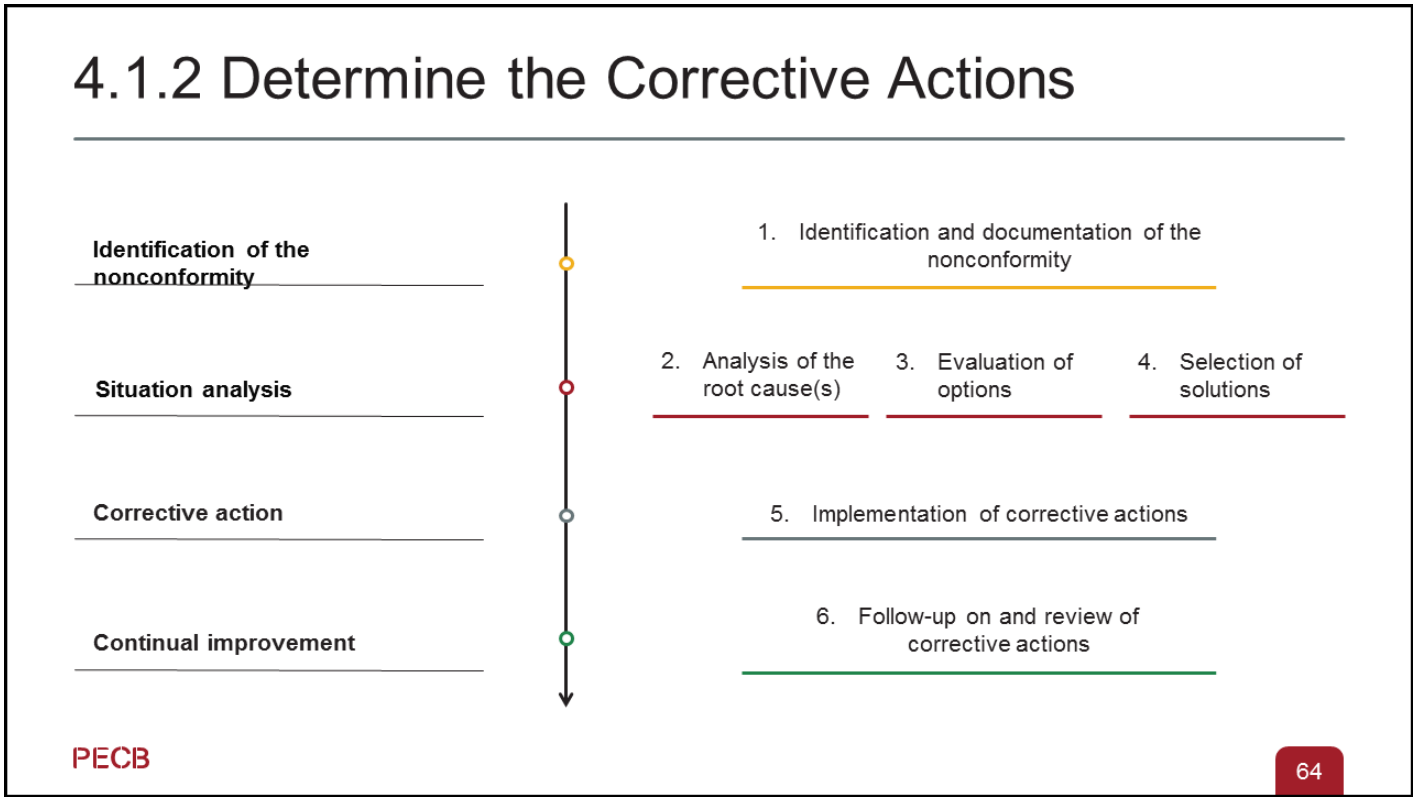

A corrective action is an action taken to permanently eliminate the root cause(s) of a nonconformity or of any other existing undesirable event and to prevent its reoccurrence. A corrective action is, thus, a term that includes the reaction to a problem related to a system or process, security incidents, obstacles in achieving objectives, nonconformities, etc.

The corrective action process should include:

1. **Identification and documentation of the nonconformity:** the person(s) responsible define and document the nonconformity and analyze its impact on the organization.
2. **Analysis of the root cause(s):** the person(s) responsible determine the source of the nonconformity and analyze its root cause(s).
3. **Evaluation of options:** the possible corrective actions are listed. At this stage, if the problem is significant, or if the probability that the problem will reoccur is high, temporary corrective actions can be implemented.
4. **Selection of solutions:** one or more corrective actions are selected to correct the problem and the improvement objectives are determined. The selected solution must correct the problem and should also ensure that the problem will not reoccur.
5. **Implementation of corrective actions:** the approved corrective action plan is implemented and all the taken actions described in the plan are documented.
6. **Follow-up on and review of corrective actions:** the person(s) responsible must monitor the effectiveness of the implanted corrective actions. This can be done by periodically assessing whether the organization is achieving its information security objectives through the corrective actions implemented.

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4.1.3 Determine the Preventive Actions

- A preventive action is any action taken to eliminate the occurrence of potential cause(s) of nonconformity or any other undesirable event in the future.
- The resources and time allocated to the preventive action should correspond with the severity of the potential nonconformity that it is trying to prevent.
- In cases where potentially high disruption nonconformities are well-known, preventive action is usually more cost-effective (e.g., implementing a firewall is more effective than dealing with a cyberattack).

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By establishing a continual risk management process, the organization is more likely to detect a change in the risk factors that concern the organization. Information security threats and vulnerabilities can change abruptly, so the organization should continuously monitor them to detect these changes and take actions to prevent information security incidents.

To determine if there is a need for preventive actions, the organization should, among others, monitor the following:

- New assets in the ISMS
- Modifications to the value of assets, for example, because of the evolution in operational needs
- New threats (internal or external) that have not been evaluated
- New vulnerabilities identified that have not been evaluated
- Identified vulnerabilities to determine those exposed to new threats
- Security incidents

Similarly to the corrective action process, a preventive action process includes: identifying a potential problem, evaluating options, selecting solutions, implementing preventive actions, and reviewing preventive actions.

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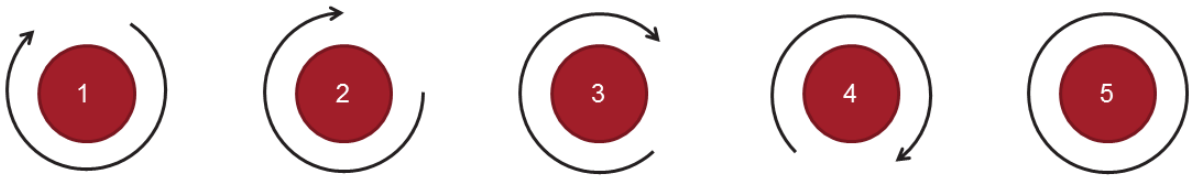
4.1.4 Draft an Action Plan

An action plan:



1. Can be written in a summarized fashion
2. Must accomplish the correction of the nonconformity
3. Should be based on a preventive and corrective approach
4. Must include an implementation time frame
5. Must produce verifiable results

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Implementation dates must be realistic and based on the nonconformities observed and the costs of the corrective measures to be taken. The deadlines set must be reasonable.

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Submission of Action Plans Following an Audit

- An action plan must be submitted to address each nonconformity separately. A general action plan devised to address all nonconformities is unacceptable.
- Action plans must be approved by the top management.
- The auditor analyzes the cause(s) and evaluates if the specific correction and corrective actions taken or planned can eliminate the detected nonconformities within a defined timeframe.

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The top management may decide to accept the risk instead of implementing corrective, preventive, or improvement actions; however, they must document the justification for their decision.

The action plans must be submitted within specified deadlines. Most certification bodies (in the case of a certification audit) set a deadline between 10 and 60 days for the submission of action plans. If the action plan is not received within the specified time period, the auditee will not be recommended for certification.

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Action Plans

Example

1. A new data system must be installed to manage clients' accounts in the network, and to separate the confidential data from other databases (time frame: within three weeks).

2. A new version of the information security policy must be published to include legal and regulatory statements, and contract requirements (time frame: within two months).

3. Establish and communicate a formal disciplinary process to take action against employees and other relevant interested parties who have committed an information security policy violation (time frame: within three months).

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Section Summary:

- A corrective action is an action taken to permanently eliminate the root cause(s) of a nonconformity or of any other existing undesirable event and to prevent its reoccurrence.
- A preventive action is any action taken to eliminate a potential nonconformity or any other undesirable event and to prevent their occurrence in the future.
- An action plan must be submitted to address each nonconformity separately. A general action plan to address all nonconformities is unacceptable.
- Action plans must be approved by the top management and they must have a time frame.

Questions?

Quiz 25

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Section 25

Continual improvement

- Continual monitoring of change factors
- Maintenance and improvement of the ISMS
- Continual update of the documented information
- Documentation of the improvements

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This section provides information that will help participants gain knowledge on the continual improvement of the information security management system through the monitoring of change factors, the update of documented information, the maintenance and improvement of the ISMS, and so on.

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Continual Improvement

1. Define and establish		2. Implement and operate		3. Monitor and review		4. Maintain and improve	
1.1	Understanding the organization and its context	2.1	Selection and design of controls	3.1	Monitoring, measurement, analysis, and evaluation	4.1	Treatment of nonconformities
1.2	ISMS scope	2.2	Implementation of controls	3.2	Internal audit	4.2	**Continual improvement**
1.3	Leadership and project approval	2.3	Management of documented information	3.3	Management review		
1.4	Organizational structure	2.4	Communication				
1.5	Analysis of the existing system	2.5	Competence and awareness				
1.6	Information security policy	2.6	Management of security operations				
1.7	Risk management						
1.8	Statement of Applicability						

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ISO/IEC 27001's Requirements for Continual Improvement

ISO/IEC 27001, clause 10.1

The organization shall continually improve the suitability, adequacy and effectiveness of the information security management system.

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ISO/IEC 27003, clause 10.2 Continual improvement

Explanation

A systematic approach using continual improvement will lead to a more effective ISMS, which will improve the organization's information security. Information security management leads the organization's operational activities in order to avoid being too reactive, i.e. that most of the resources are used for finding problems and addressing these problems. The ISMS is working systematically through continual improvement so that the organization can have a more proactive approach. Top management can set objectives for continual improvement, e.g. through measurements of effectiveness, cost, or process maturity.

Guidance

Continual improvement of the ISMS should entail that the ISMS itself and all of its elements are assessed considering internal and external issues, requirements of the interested parties and results of performance evaluation. The assessment should include an analysis of:

a. *suitability of the ISMS, considering if the external and internal issues, requirements of the interested parties, established information security objectives and identified information security risks are properly addressed through planning and implementation of the ISMS and information security controls;*
b. *adequacy of the ISMS, considering if the ISMS processes and information security controls are compatible with the organization's overall purposes, activities and processes; and*
c. *effectiveness of the ISMS, considering if the intended outcome(s) of the ISMS are achieved, the requirements of the interested parties are met, information security risks are managed to meet information security objectives, nonconformities are managed, while resources needed for the establishment, implementation, maintenance and continual improvement of the ISMS are commensurate with those results.*

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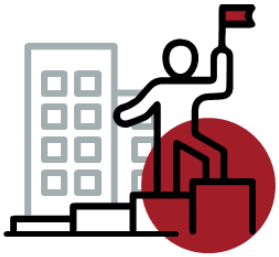
Continual Improvement

Continual improvement is the process of increasing the effectiveness and efficiency of the organization to fulfill its policy and objectives.



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Continual improvement is achieved by setting organizational performance goals, measuring and reviewing, and making necessary modifications in processes, systems, resources, etc.

Another way for an organization to demonstrate that it continuously strives to improve its ISMS is by publishing their performance. This can be done in two ways.

1. Informing the employees of what has been achieved (e.g., monthly): The benefits are manifold: employees become aware of where the organization stands in the market; they can see the result of their contribution and better tie their personal goals to organization's growth.
2. Publishing a report on organizational performance over a period of time (e.g., annually): This practice allows organizations to present their accomplishments to the customer and other interested parties.

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4.2 Continual Improvement

List of activities

- **4.2.1** Monitor change factors
- **4.2.2** Maintain and improve the ISMS
- **4.2.3** Maintain and update documented information
- **4.2.4** Document the improvements

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4.2.1 Monitor Change Factors

ISMS change factors to monitor:

Organizational changes	Changes in technologies	External changes	Changes from the ISMS
• Mission • Business objectives • Budget and resources • New products and services • Employee turnover	• Hardware • Software • IT procedures • IT processes	• Laws and regulations • Concerns and requirements of clients or suppliers • Vendors • New competitors	• Information security policy • New risk scenarios • Changes in procedures • Results of testing and exercises • Audit results

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To be effective, the management system should accurately reflect the business requirements, organizational structure, and internal policies and procedures.

As part of the continual improvement process, activities and procedures undergo changes because of the changes in business needs, technology advancements, or new internal and external policies. Therefore, it is essential that the management system is reviewed and updated regularly as part of the organization's change management process to ensure that the new information is documented and controls are revised, if required.

As a general rule, the plan should be reviewed for accuracy and completeness at a defined frequency or whenever significant changes occur to any element of the plan. Some elements may require more frequent reviews.

While all strategies should be reviewed on an ongoing basis, the frequency that an organization's ISMS should be reviewed depends on the nature, scale, and complexity of the organization, its business risk profile, and the environment in which it operates. Good practice indicates that the organization's strategy should be reviewed at least every 12 months, unless:

- It is the initial development and documented evidence of the strategy.
- It undergoes a significant change in the key technology, including systems or networks.
- The pace of business change is particularly aggressive.

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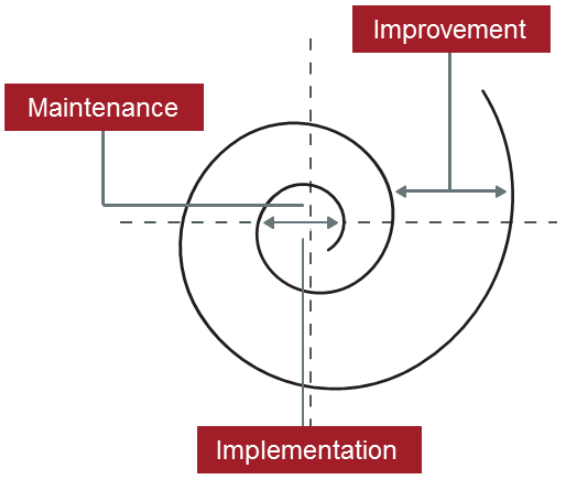
4.2.2 Maintain and Improve the ISMS



- The ISMS needs to be maintained and updated periodically.
- The information security manager should be notified regarding any agreed actions to be taken to improve the processes so that no risk or risk element is overlooked or underestimated before the changes are implemented.

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4.2.3 Maintain and Update Documented Information

Examples

ISMS documentation

- Information security policy
- Information security objectives and targets
- Awareness and training programs
- Risk analysis
- Documented information control procedure
- Incident management plans
- Internal audit program
- Management review plans

Factors of change

- Organizational changes
- New rules
- Changes in the business scope
- Incidents
- Faulty operations
- Failures
- Risk management reports
- Test results
- Internal and external audits reports

Review and update

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Documented information is the cornerstone of the proper functioning of the information security management system. In the event of a crisis, it is important to have complete and up-to-date documented information in order to allow the actors involved to follow a guide of actions instead of taking decisions based on improvisation or intuition.

The proper maintenance of documented information eliminates to a great degree the possibility of taking spontaneous decisions. In addition, it makes the principal actors ready to act when the situation requires it, giving guidance, and avoiding as many mistakes as possible.

The ISMS is a dynamic system and continual change is imperative. As a consequence, the ISMS documented information must be adapted on each and every trigger of change.

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The Benefits of Continual Improvement

Increased efficiency

Continual improvement increases productivity as the changes may lead to long-term positive outputs.

Collaborative teams

Working continuously toward a common goal will help in building new relationships within teams and reinforcing the existing ones.

Increased customer satisfaction

While actively improving processes, organizations simultaneously increase the quality of the products and services that they offer.

Error reduction

With the continual improvement of processes, the number of errors in those processes will also be reduced.

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4.2.4 Document the Improvements

It is good practice to document continual improvement activities in a register. This register can take the form of a spreadsheet or a database in which activities are collected, prioritized, and tracked to completion.

Continual Improvement Register					
No.	Change description	Priority	Date of initiation	Date of completion	Signature

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The improvements or modifications made to the ISMS should be recorded by using a record of changes, which lists a special identifier (e.g., change number or any specific code), description of the change being made, priority (which can be classified as high, medium, low or based on numbers of 1 to 5), dates of change initiation and completion, and the signature of the person overseeing the change.

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Section Summary:

- The organization must continually improve the suitability, adequacy, and effectiveness of the ISMS.
- Continual improvement refers to the process of increasing the effectiveness and efficiency of the organization to fulfill its policy and objectives.
- The ISMS needs to be maintained and updated periodically.
- Continual improvement helps organizations fulfill their policies and objectives.
- The organization should maintain and update its ISMS documentation, including the information security policy, information security objectives, awareness and training programs, ISMS risk analyses, incident management plans, management review plans, and the internal audit program.
- Continual improvement helps organizations increase efficiency and customer satisfaction, reduce errors, and build teamwork.

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Questions?

Quiz 26

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This section provides information that will help participants gain knowledge about accreditation and certification bodies. In addition, participants will be able to understand the certification process, including the selection of the certification body, the stage 1 and 2 audit, the audit follow-up, the certification recommendation and decision, and the recertification audit.

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Accreditation Bodies

- An accreditation body is an authoritative, independent organization that verifies whether a conformity assessment body meets established criteria and is competent to carry out conformity assessment tasks.
- Activities covered by accreditation include but are not limited to: testing, calibration, inspection, certification of management systems, persons, products, processes and services, and validation and verification.
- Accreditation bodies usually have their authority from government.

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ISO/IEC 17011 provides general requirements for accreditation bodies in assessing and accrediting certification bodies. Compliance with the requirements of ISO/IEC 17011 proves that the accreditation bodies are competent and reliable in offering accreditation services.

Commonly, there is only one accreditation authority in each country. However, in the United States, there are several accreditation bodies: IAS and ANAB.

- The **International Accreditation Service (IAS)** accredits certification programs for persons, products, and management systems according to ISO/IEC 17024, ISO/IEC 17065, and ISO/IEC 17021-1.
- The **ANSI National Accreditation Board (ANAB)** supervises the certification bodies accredited against ISO/IEC 17021-1.

Accreditation authority groups:

- **European co-operation for Accreditation (EA)** is the European network of accreditation organizations based in Europe. The members include UKAS, COFRAC, BNAC, ENAC, etc.

Source: https://european-accreditation.org

- **International Accreditation Forum (IAF)** is the international association of accreditation organizations for systems in management, product, services, individuals, and other programs. The objective of IAF is to ensure that the member organizations only certify competent organizations and establish agreements of mutual recognition among its members.

Source: https://www.iaf.nu

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The following is a list of accreditation authorities for several countries (see the complete list on the IAF website: https://www.iaf.nu):

Argentina: Organismo Argentino de Acreditación (OAA), www.oaa.org.ar

Australia and New Zealand: Joint Accreditation System of Australia and New Zealand (JAS-ANZ), www.jas-anz.org

Austria: Federal Ministry of Economy, Family and Youth (BMWFJ), www.en.bmdw.gv.at

Belgium: Belgian Accreditation Body (BELAC), www.belac.fgov.be

Brazil: General Coordination for Accreditation (CGCRE), www.inmetro.gov.br

Canada: Standards Council of Canada (SCC), www.scc.ca

Chile: Instituto Nacional de Normalizacion (INN), www.inn.cl

China: China National Accreditation Service for Conformity Assessment (CNAS), https://www.cnas.org.cn/english/index.shtml

Egypt: Egyptian Accreditation Council (EGAC), www.egac.gov.eg

Finland: Finnish Accreditation Service (FINAS), www.finas.fi

France: Comité Français d'Accréditation (COFRAC), www.cofrac.fr

Germany: Deutsche Akkreditierungsstelle GmbH (DAkkS), www.dakks.de

Hong Kong, China: Hong Kong Accreditation Service (HKAS), www.itc.gov.hk/hkas

India: National Accreditation Board for Certification Bodies (NABCB), www.qcin.org

Iran: National Accreditation Center of Iran (NACI), http://www.naci.isiri.org

Ireland: Irish National Accreditation Board (INAB), www.inab.ie

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Japan: International Accreditation Japan (IAJapan), www.jab.or.jp

Korea: Korea Accreditation System (KAS), www.iaf.nu/articles/IAF_MEM_Korea_Republic_of_/86

Malaysia: Standards Malaysia (DSM), www.jsm.gov.my

Mexico: Mexican Accreditation Entity (EMA), www.ema.org.mx

Netherlands: Dutch Accreditation Council (Raad Voor Accreditatie) (RvA), www.rva.nl

Norway: Norwegian Accreditation (NA), www.akkreditert.no

Pakistan: Pakistan National Accreditation Council (PNAC), www.pnac.org.pk

Philippines: Philippine Accreditation Office (PAB), www.dti.gov.ph

Portugal: Portuguese Institute for Accreditation (IPAC), www.ipac.pt

Romania: Romanian Accreditation Association (RENAR), www.renar.ro

Russian Federation: Scientific Technical Center on Industrial Safety (STC-IS), www.oaontc.ru/en/

Singapore: Singapore Accreditation Council (SAC), www.sac-accreditation.gov.sg

Slovenia: Slovenska Akreditacija (SA), www.slo-akreditacija.si

South Africa: South African National Accreditation System (SANAS), www.sanas.co.za

Spain: Entidad Nacional de Acreditacion (ENAC), www.enac.es

Sweden: Swedish Board for Accreditation and Conformity Assessment (SWEDAC), www.swedac.se/?lang=en

Switzerland: State Secretariat for Economic Affairs, Swiss Accreditation Service (SAS), www.sas.ch

Thailand: National Standardization Council of Thailand (NSC), www.tisi.go.th

Tunisia: Tunisian Accreditation Council (TUNAC), www.tunac.tn

Turkey: Turkish Accreditation Agency (TURKAK), www.turkak.org.tr

United Arab Emirates: Emirates International Accreditation Center (EIAC), www.eiac.gov.ae

United Kingdom: United Kingdom Accreditation Service (UKAS), www.ukas.com

United States: ANSI-ASQ National Accreditation Board (ANAB), www.anab.org

United States: American National Standards Institute (ANSI), www.ansi.org

United States: International Accreditation Services (IAS), www.iasonline.org

Uruguay: Organismo Uruguayo de Acreditacion (OUA)

Vietnam: Bureau of Accreditation (BoA), www.boa.gov.vn/en

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Certification Bodies

Certification bodies certify management systems (ISO/IEC 17021-1), persons (ISO/IEC 17024), and products, processes, and services (ISO/IEC 17065).

Certification bodies are always third-party, impartial conformity assessment bodies.

A certification body can be a governmental or nongovernmental organization, with or without regulatory authority.

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ISO/IEC 17021-1, Introduction

Certification of a management system provides independent demonstration that the management system of the organization:

a. *conforms to specified requirements;*
b. *is capable of consistently achieving its stated policy and objectives;*
c. *is effectively implemented.*

ISO/IEC 17024, Introduction

This International Standard has been developed with the objective of achieving and promoting a globally accepted benchmark for organizations operating certification of persons. Certification for persons is one means of providing assurance that the certified person meets the requirements of the certification scheme.

In either case, this International Standard can serve as the basis for the recognition of the certification bodies for persons and the certification schemes under which persons are certified, in order to facilitate their acceptance at the national and international levels.

ISO/IEC 17065, Introduction

The overall aim of certifying products, processes or services is to give confidence to all interested parties that a product, process or service fulfils specified requirements.

Parties that have an interest in certification include, but are not limited to:

a. *the clients of the certification bodies;*
b. *the customers of the organizations whose products, processes or services are certified;*
c. *governmental authorities;*
d. *non-governmental organizations; and*
e. *consumers and other members of the public.*

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Management System Certification Bodies

- Management system certification bodies should conduct their activities in a competent, consistent, and impartial manner.
- For this purpose, ISO/IEC 17021-1 sets out the requirements that these bodies must adhere. Requirements include:
  - General requirements
  - Structural requirements
  - Resource requirements
  - Information requirements
  - Process requirements
  - Management system requirements
- Adherence to ISO/IEC 17021-1 not only facilitates the recognition of a certification body, it also ensures the acceptance of their certifications on a national and international basis.

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Apart from the aforementioned requirements, ISO/IEC 17021-1 acknowledges the fact that varying competences are needed for auditing different management systems. This acknowledgement is reflected in the fact that ISO has developed a series of parts of the main ISO/IEC 17021 standard that deal with auditor competence for different types of management systems.

The following standards specify competence requirements for auditing and certification in the following areas:

- ISO/IEC 17021-2: Environmental management systems
- ISO/IEC 17021-3: Quality management systems
- ISO/IEC TS 17021-4: Event sustainability management systems
- ISO/IEC TS 17021-5: Asset management systems
- ISO/IEC TS 17021-6: Business continuity management systems
- ISO/IEC TS 17021-7: Road traffic safety management systems
- ISO/IEC TS 17021-8: Management systems for sustainable development in communities
- ISO/IEC TS 17021-9: Anti-bribery management systems
- ISO/IEC TS 17021-10: Occupational health and safety management systems
- ISO/IEC TS 17021-11: Facility management systems
- ISO/IEC TS 17021-12: Collaborative business relationship management systems

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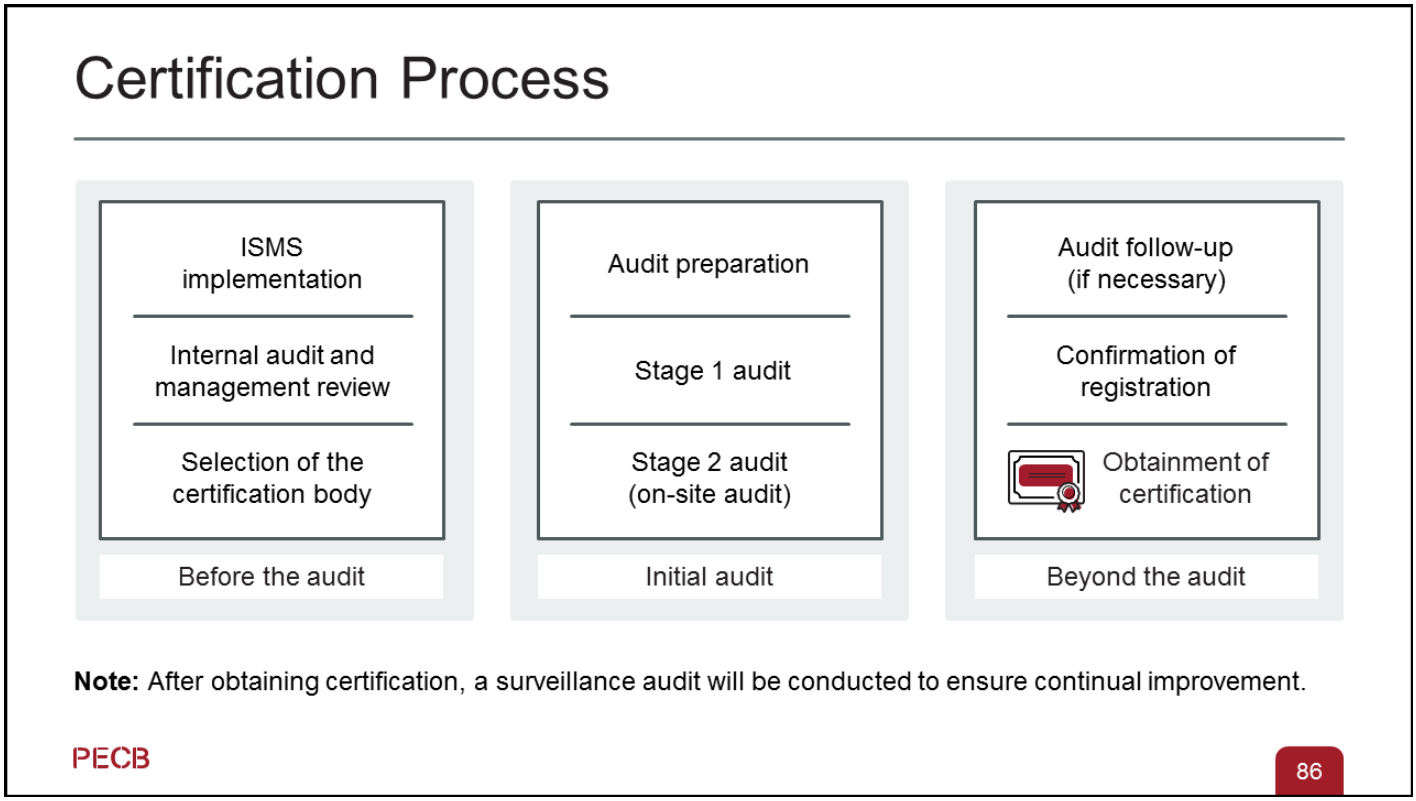

Note:

- Continual improvement refers to the ongoing process that an organization undergoes in order to improve their procedures, processes, and products or services.
- Surveillance audit refers to the activity that is performed once a year (sometimes more, based on the organization's needs) to ensure that their management system is in conformity to the respective management system standard.

1. **Selection of the certification body (registrar):** Organizations are free to select the certification body. When doing so, they should keep in mind that more renowned certification bodies offer greater recognition.
2. **Pre-assessment audit (optional):** Pre-assessment audit allows organizations to measure any gaps between their existing management system and the requirements of the standard.
3. **Stage 1 audit:** The main objective of stage 1 audit is to verify whether the management system is designed to meet the requirements of the standard and the objectives of the organization. It is recommended to perform at least some portion of the stage 1 audit on-site (at the organization's premises).
4. **Stage 2 audit (on-site visit):** The objective of stage 2 audit is to evaluate whether the management system conforms to all the requirements of the standard, is being implemented, and can support the organization in achieving its objectives. Stage 2 audit is conducted at the premises of the organization.
5. **Audit follow-up:** If nonconformities have been detected, the auditor will perform a follow-up visit to validate only the action plans associated with those nonconformities (which usually takes up to one day).
6. **Confirmation of registration:** If the organization complies with the requirements of the standard, the certification body confirms the registration and publishes the certificate.

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Certification and Attestation

Differences

Certification

Certification is a formal procedure which attests to a status or a level of achievement by providing an official document.

Attestation

Attestation is a method used to check, confirm, and authenticate the validity of a document.

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Selecting the Certification Body

The following are the main criteria in selecting a certification body:

1. Reputation and credibility
2. Geographical location
3. References in your sector
4. Possibility of a combined audit
5. Skills and experience of the audit team
6. Prices

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1. **Reputation and credibility:** The value of certification depends on the reputation and credibility of the certification body that issues the certificate. As a result, it is important to select a credible certification body.
2. **Geographical location:** Choosing a certification body that operates in your area or that the audit team members speak the local language and are familiar with the local customs is advisable.
3. **References in your sector:** If the industry you operate in has specific regulatory requirements, selecting a certification body that already has clients in your business sector is preferable.
4. **Possibility of a combined audit:** If you consider certifying your organization against several standards (e.g., ISO 9001 and ISO 14001), you may want to ensure that the certification body can provide combined audits.
5. **Skills and experience of the audit team:** It is best practice to contact the certification body to ensure that the audit team has the necessary competencies and skills to perform the audit.
6. **Prices:** Prices vary between certification bodies, but you may want to request a few bids, as the number of days per audit proposed by the certification body may differ, influencing audit costs.

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Audit Time Calculation

ISO/IEC 27006, Table B.1 (excerpt)

Number of employees	***Audit time*** *(day/person) ISO/IEC 27001*
1 to 10	*5*
11 to 15	*6*
16 to 25	*7*
26 to 45	*8.5*
46 to 65	*10*
66 to 85	*11*
86 to 125	*12*
126 to 175	*13*
176 to 275	*14*

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Audit Time Calculation (Cont'd)

ISO/IEC 27006, Table B.1 (excerpt)

Number of employees	***Audit time*** *(day/person) ISO/IEC 27001*
276 to 425	*15*
426 to 625	*16.5*
626 to 875	*17.5*
876 to 1175	*18.5*
1176 to 1550	*19.5*
1551 to 2025	*21*
2026 to 2675	*22*
2676 to 3450	*23*
3451 to 4350	*24*

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Certification bodies must give auditors enough time to complete the audit. The time available to complete an audit can vary depending on the following:

- Scope of the management system
- Complexity of the processes of the management system
- Field of activity of the auditee
- Complexity and diversity of the technologies in use
- Number of sites to audit
- Previous audits
- Agreements related to outsourced services
- Regulations, laws, and contract agreements

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Rejection of an Auditor by the Audit Client or Auditee

ISO/IEC 17021-1, clause 9.2.3.5

The certification body shall provide the name of and, when requested, make available background information on each member of the audit team, with sufficient time for the client to object to the appointment of any particular audit team member and for the certification body to reconstitute the team in response to any valid objection.

Examples of valid reasons for which the auditee can reject an auditor:

- The auditor is in a conflict of interest situation (real or potential).
- The auditor has previously displayed unprofessional conduct.
- The auditor does not hold the security clearance required by the auditee.
- The auditor has audited the organization in the past, as such the audit client is not confident that they will receive added value from the audit.

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The audit client or the auditee can request the replacement of an audit team member for valid reasons. A valid reason would be the case of an auditor having previously displayed unprofessional conduct. Other examples of valid reasons are situations with real conflict of interest (an auditor has previously worked for the auditee) or perceived conflict of interest (an auditor has worked for one of the auditee's major competitors).

In some industries and sectors (arms industry, nuclear power, information services, etc.), an auditee can request that each member of the audit team holds a security clearance or that a background check on each member is conducted before being admitted on-site.

It is recommended to communicate these reasons to the persons responsible for the audit team and to the persons responsible for the audit program before requesting the replacement of an audit team member.

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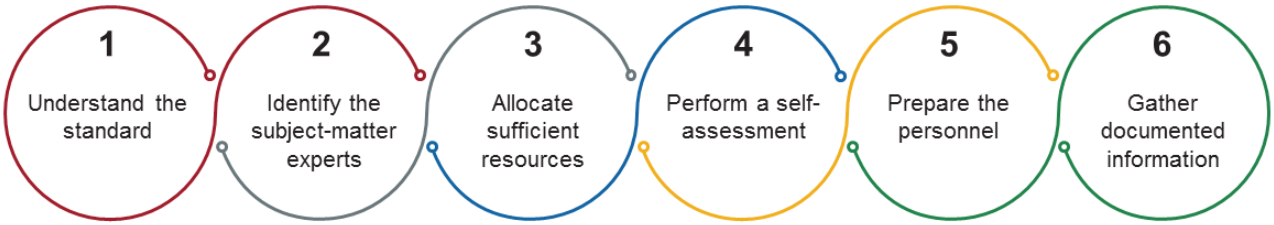
Preparing for the Certification Audit

Recommendations



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To be well prepared for a certification audit, the organization can take the following actions prior to the commencement of the audit:

1. **Understand the standard:** The whole audit will center around the standard against which the organization is seeking certification. A general understanding of the standard can help the organization efficiently manage the external audit.
2. **Identify the subject-matter experts:** The organization must determine which of their employees have the knowledge to help the external auditor understand and evaluate the processes. If the organization has already assigned a person or team to manage the ISMS, this step is unnecessary.
3. **Allocate sufficient resources:** An external audit requires time and effort from the organization.
4. **Perform a self-assessment:** The organization should review its compliance with the requirements of ISO/IEC 27001 and address the following questions:
    - Are the processes clearly defined?
    - Have the roles and responsibilities been segregated?
    - Do we maintain documented information?
    - Are the processes effective in producing the desired results?
5. **Prepare the personnel:** The organization should prepare its employees for the external audit by:
    - Informing them about the audit
    - Organizing training sessions
6. **Gather documented information:** Because external auditors will ask for evidence that procedures are in place, it is preferable to prepare documented information in advance so as to avoid losing time.

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Stage 1 Audit

1

Site visit

- Assess the auditee's location and site-specific conditions
- Meet with the auditee's personnel
- Observe the technologies used
- Observe the operations of the ISMS

2

Interviews with key interested parties

- Validate the ISMS scope as well as the applicable legal, regulatory, and contractual constraints
- Verify if procedures and policies are followed
- Prepare for stage 2 audit

3

Documented information review

- Understand how the ISMS operates
- Evaluate the design of the ISMS as well as its related processes or controls
- Verify that internal audits and management reviews have been performed

Note: The review of documented information is the main activity of stage 1 audit.

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During the stage 1 audit, the auditor does not verify the effectiveness of the management system, but rather the design of the management system. The auditor will verify the effectiveness of the management system during the stage 2 audit (on-site audit) to validate whether the documented processes exist, are effective, and comply with the standard requirements.

Usually, 30% of the total time is spent on the stage 1 audit.

The stage 1 audit should not be conducted too far from the stage 2 audit, so that the management system does not change substantially between the two stages. It should, however, be conducted far enough apart to prepare the on-site audit plan. The stage 1 audit ideally takes place two to four weeks before the stage 2 audit (on-site).

It is noteworthy that although a confidentiality agreement is signed, the auditee has the right to require that the documented information review takes place on-site and that no documented information is carried off-site.

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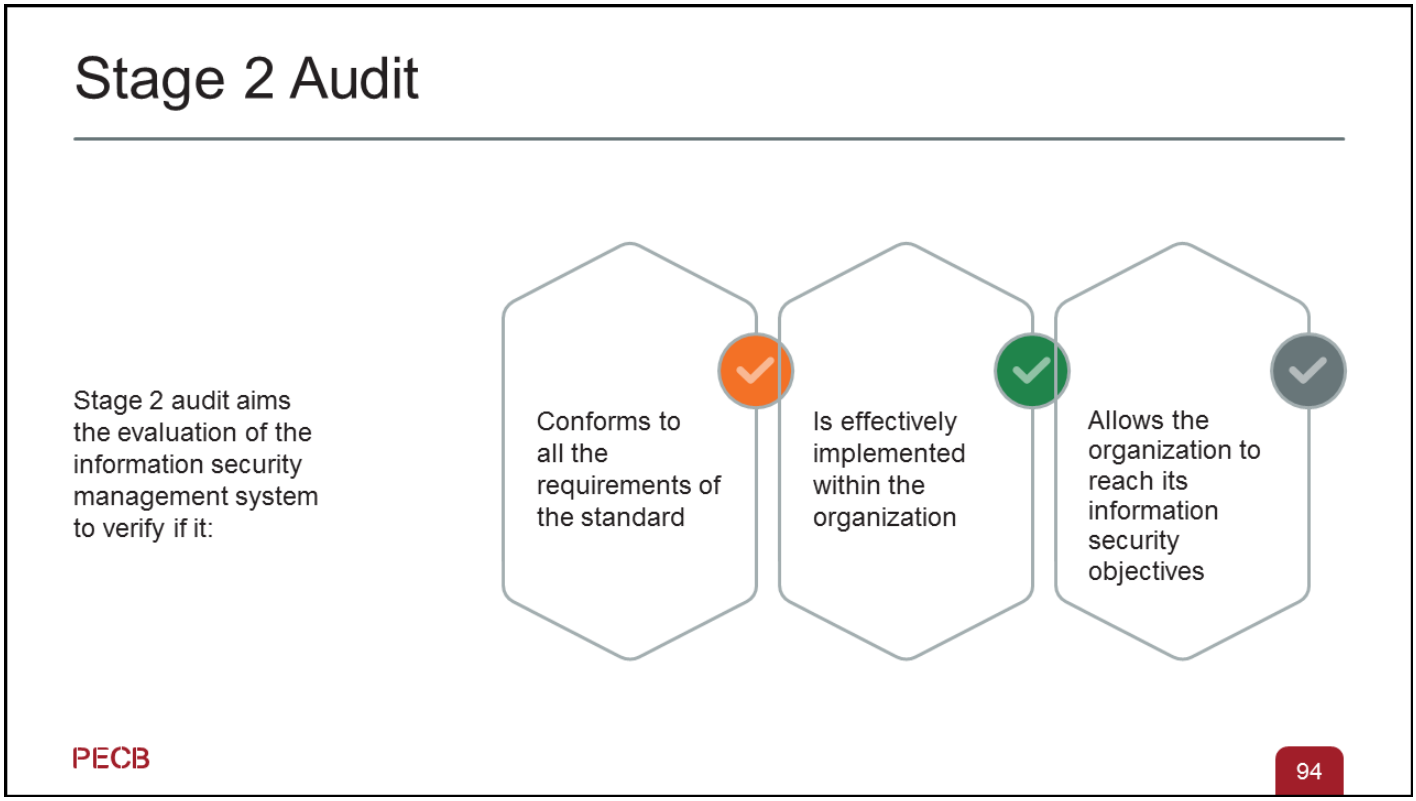

ISO/IEC 17021-1, clause 9.3.1.3 Stage 2

The purpose of stage 2 is to evaluate the implementation, including effectiveness, of the client's management system. The stage 2 shall take place at the site(s) of the client. It shall include the auditing of at least the following:

a. *information and evidence about conformity to all requirements of the applicable management system standard or other normative documents;*
b. *performance monitoring, measuring, reporting and reviewing against key performance objectives and targets (consistent with the expectations in the applicable management system standard or other normative document);*
c. *the client's management system ability and its performance regarding meeting of applicable statutory, regulatory and contractual requirements;*
d. *operational control of the client's processes;*
e. *internal auditing and management review;*
f. *management responsibility for the client's policies.*

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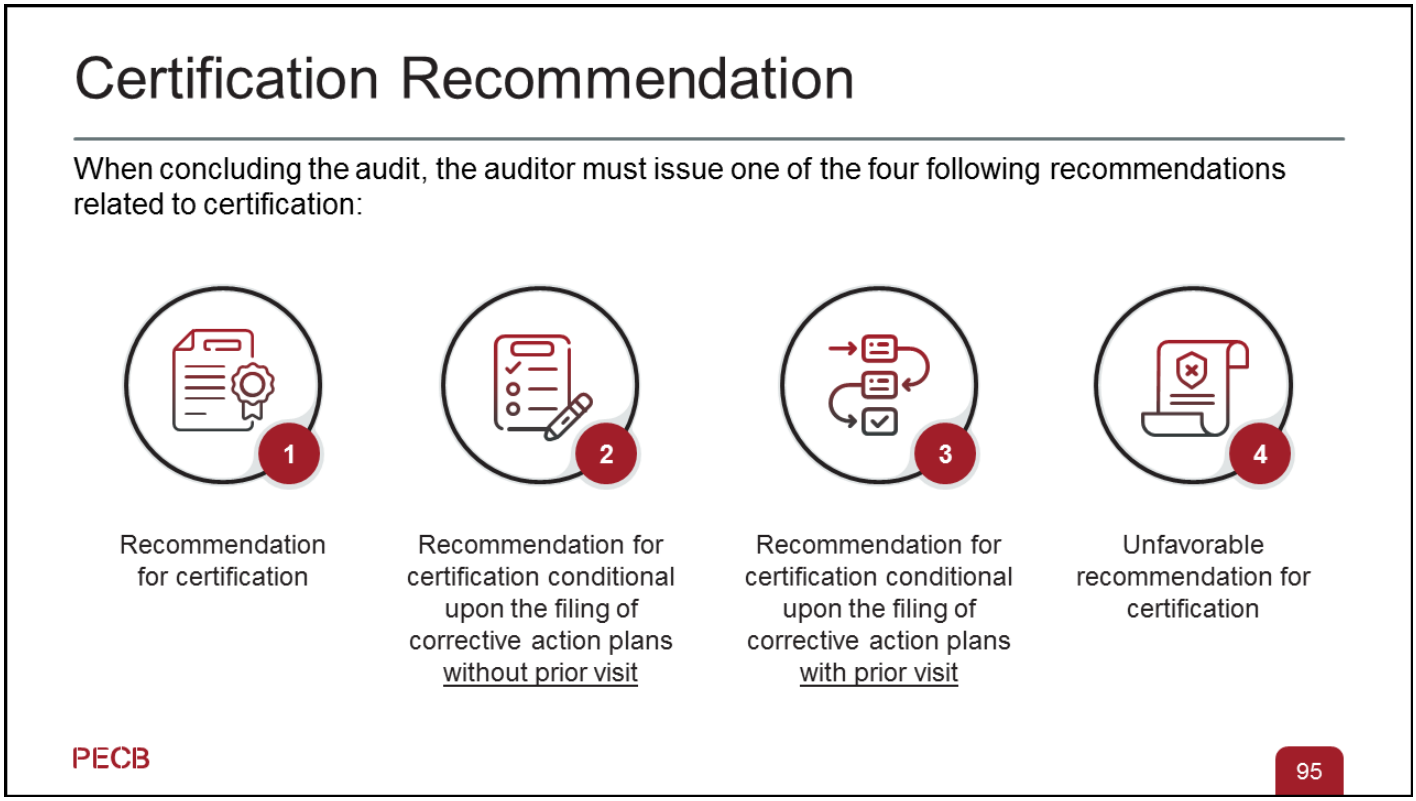

1.Recommendation for certification: Based on the evidence gathered during the audit, the auditor is confident that the auditee is in conformity to the requirements of the standard. The auditor did not detect any nonconformity during the audit.

2.and 3. Recommendation for certification conditional upon the filing of corrective action plans: The auditor is confident that the auditee is in conformity to the requirements of the standard. However, the auditor detected a few minor nonconformities. As such, the auditee is required to submit corrective actions plans for each minor nonconformity within a short period of time. If the corrective action plans are accepted, the auditee can then be certified. There are some cases when the auditor requires a new on-site visit prior to issuing a favorable certification recommendation. If there is no additional on-site visit required, the corrective action plans will be verified and validated during the surveillance audit visits.

4.Unfavorable recommendation for certification: The auditor is confident that the auditee is not in conformity to the requirements of the standard. The auditor detected one or more major nonconformities during the audit. As such, the auditor issues an unfavorable recommendation. It is worth mentioning that there are no public statements made about organizations that received unfavorable recommendation. A public statement is made for organizations that received favorable recommendation (except in some cases).

Please note that the auditor only issues a recommendation for certification. The final certification decision is made by the certification committee of the certification body.

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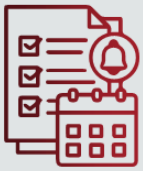
Audit Follow-up

Based on the audit conclusions, the auditor may have to conduct an audit follow-up before the organization is recommended for certification.

During the audit follow-up, the auditor evaluates the effectiveness of all corrections and corrective actions taken.



If a major nonconformity is detected and reported, then the auditee will be subject to audit follow-up.

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The audit follow-up activities should be planned carefully and in detail just as the other steps included in the conduct of the audit.

The audit follow-up's objective is to validate the actions plans and the implemented corrective actions submitted by the auditee. If any major nonconformity is detected, the auditee must resolve them before being recommended for certification.

The audit follow-up is usually conducted 4 to 12 weeks after the initial audit so that the auditee is given time to implement the corrective actions. The conduct of audit follow-up is usually completed within a day.

ISO 19011, clause 6.7 Conducting audit follow-up

The outcome of the audit can, depending on the audit objectives, indicate the need for corrections, or for corrective actions, or opportunities for improvement. Such actions are usually decided and undertaken by the auditee within an agreed timeframe. As appropriate, the auditee should keep the individual(s) managing the audit program and/or the audit team informed of the status of these actions.

The completion and effectiveness of these actions should be verified. This verification may be part of a subsequent audit. Outcomes should be reported to the individual managing the audit program and reported to the audit client for management review.

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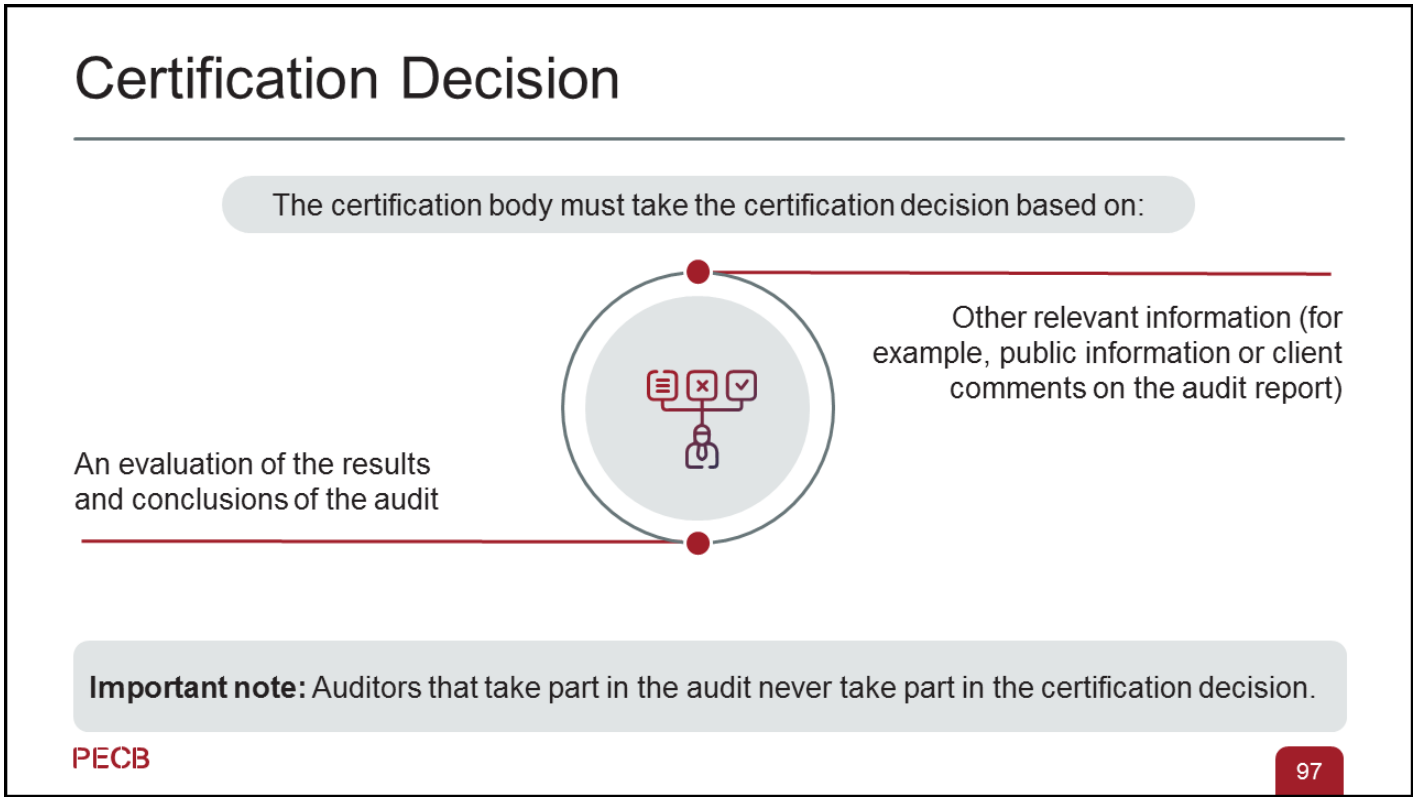

ISO/IEC 17021-1, clause 9.5.1.1

The certification body shall ensure that the persons or committees that make the decisions for granting or refusing certification, expanding or reducing the scope of certification, suspending or restoring certification, withdrawing certification or renewing certification are different from those who carried out the audits. The individual(s) appointed to conduct the certification decision shall have appropriate competence.

ISO/IEC 17021-1, clause 9.5.3.1

The information provided by the audit team to the certification body for the certification decision shall include, as a minimum:

a. *the audit report;*
b. *comments on the nonconformities and, where applicable, the correction and corrective actions taken by the client;*
c. *confirmation of the information provided to the certification body used in the application review;*
d. *confirmation that the audit objectives have been achieved;*
e. *a recommendation whether or not to grant certification, together with any conditions or observations.*

ISO/IEC 17021-1, clause 9.5.3.2

If the certification body is not able to verify the implementation of corrections and corrective actions of any major nonconformity within 6 months after the last day of stage 2, the certification body shall conduct another stage 2 prior to recommending certification.

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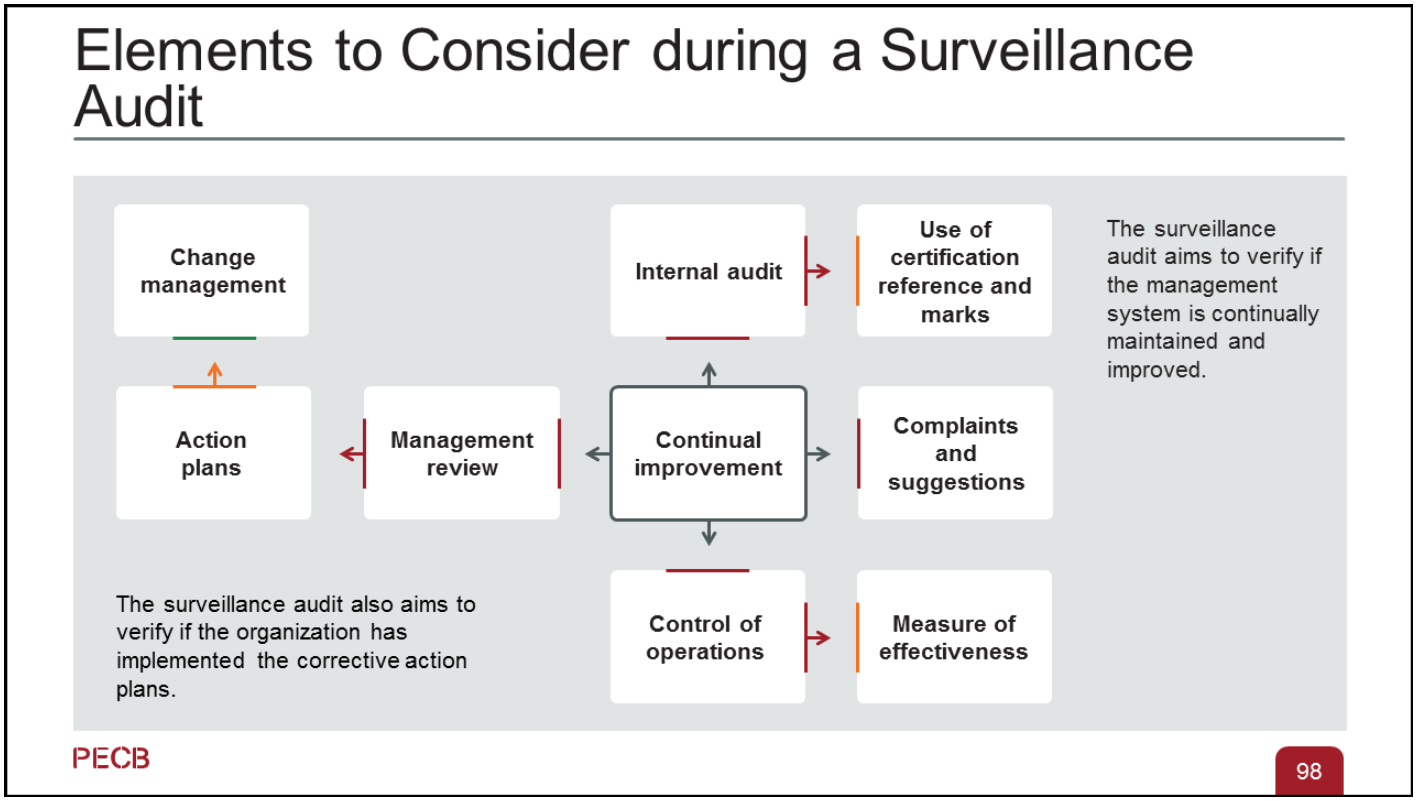

ISO/IEC 17021-1, clause 9.6.2.2 Surveillance audit

Surveillance audits are on-site audits, but are not necessarily full system audits, and shall be planned together with the other surveillance activities so that the certification body can maintain confidence that the client's certified management system continues to fulfill requirements between recertification audits.

Each surveillance for the relevant management system standard shall include:

a. *internal audits and management review;*
b. *a review of actions taken on nonconformities identified during the previous audit;*
c. *complaints handling;*
d. *effectiveness of the management system with regard to achieving the certified client's objectives and the intended results of the respective management system (s);*
e. *progress of planned activities aimed at continual improvement;*
f. *continuing operational control;*
g. *review of any changes;*
h. *use of marks and/or any other reference to certification.*

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Recertification Audit

ISO/IEC 17021-1, clauses 9.6.3.1.1 and 9.6.3.1.2

- *The purpose of the recertification audit is to confirm the continued conformity and effectiveness of the management system as a whole, and its continued relevance and applicability for the scope of certification.*
- *A recertification audit shall be planned and conducted to evaluate the continued fulfilment of all of the requirements of the relevant management system standard or other normative document.*
- *This shall be planned and conducted in due time to enable for timely renewal before the certificate expiry date.*
- *The recertification activity shall include the review of previous surveillance audit reports and consider the performance of the management system over the most recent certification cycle.*

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ISO/IEC 17021-1, clause 9.6.3.1.3

Recertification audit activities may need to have a stage 1 in situations where there have been significant changes to the management system, the organization, or the context in which the management system is operating (e.g. changes to legislation).

NOTE Such changes can occur at any time during the certification cycle and the certification body might need to perform a special audit, which might or might not be a two-stage audit.

ISO/IEC 17021-1, clause 9.6.3.2.1

The recertification audit shall include an on-site audit that addresses the following:

a. *the effectiveness of the management system in its entirety in the light of internal and external changes and its continued relevance and applicability to the scope of certification;*
b. *demonstrated commitment to maintain the effectiveness and improvement of the management system in order to enhance overall performance;*
c. *the effectiveness of the management system with regard to achieving the certified client's objectives and the intended results of the respective management system (s).*

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Use of ISO Trademarks

- A certified auditee is authorized to publicly display its certification and use it for marketing purposes.
- The certification cannot be displayed directly on a product or in a way that would lead to believe that the product is certified.
- The certification body will provide the auditee with a logo that can be used for marketing purposes.
- The unauthorized use of ISO trademarks could mislead, create false impressions, or cause confusion. Therefore, the ISO trademarks must not be used with the intention to express the certification of a product, person, or organization since ISO does not perform certifications.

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ISO/IEC 17021-1, clause 8.3.1

A certification body shall have rules governing any management system certification mark that it authorizes certified clients to use. These rules shall ensure, among other things, traceability back to the certification body. There shall be no ambiguity, in the mark or accompanying text, as to what has been certified and which certification body has granted the certification. This mark shall not be used on a product nor product packaging nor in any other way that may be interpreted as denoting product conformity.

NOTE ISO/IEC 17030 provides additional information for use of third-party marks.

ISO/IEC 17021-1, clause 8.3.2

A certification body shall not permit its marks to be applied by certified clients to laboratory test, calibration or inspection reports or certificates.

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Section Summary:

- Certification is a formal procedure which attests to a status or a level of achievement by providing an official document, whereas attestation refers to a method used to check, confirm, and authenticate the validity of a document.
- An accreditation body is an authoritative, independent organization that verifies whether a conformity assessment body meets established criteria and is competent to carry out conformity assessment tasks.
- Certification bodies certify management systems (ISO/IEC 17021-1), persons (ISO/IEC 17024), and products, processes, and services (ISO/IEC 17065).
- A certified auditee is authorized to publicly display its certification and use it for marketing purposes.
- ISO trademarks must not be used with the intention to express the certification of a product, person, or organization, since ISO does not perform certifications.

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Questions?

Quiz 27

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Note: To complete Quiz 27, please go to the Quizzes Sheet.

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Scenario-based Quiz 4

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Note: To complete the Scenario-based Quiz 4, please go to the Quizzes Sheet.

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This section elaborates on the PECB certification scheme and process.

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PECB ISO/IEC 27001 Certification Scheme

Requirements summary

Professional credential	Exam	Professional experience	ISMS audit experience	ISMS project experience
ISO/IEC 27001 Foundation	ISO/IEC 27001 Foundation	----------	----------	----------
ISO/IEC 27001 Provisional Auditor	ISO/IEC 27001 Lead Auditor	----------	----------	----------
ISO/IEC 27001 Auditor		2 years (1 in information security management)	200 hours	----------
ISO/IEC 27001 Lead Auditor		5 years (2 in information security management)	300 hours	----------
ISO/IEC 27001 Senior Lead Auditor		10 years (7 in information security management)	1,000 hours	----------
ISO/IEC 27001 Provisional Implementer	ISO/IEC 27001 Lead Implementer	----------	----------	----------
ISO/IEC 27001 Implementer		2 years (1 in information security management)	----------	200 hours
ISO/IEC 27001 Lead Implementer		5 years (2 in information security management)	----------	300 hours
ISO/IEC 27001 Senior Lead Implementer		10 years (7 in information security management)	----------	1,000 hours
ISO/IEC 27001 Master	ISO/IEC 27001 LA + LI (four additional Foundation exams)	15 years (10 in information security management)	700 hours	700 hours

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The **"Foundation"** credential recognizes that individuals understand the basic concepts, methods, and techniques used for the effective management of a management system.

The main auditor credentials:

1. The **"Certified Provisional Auditor"** credential recognizes that individuals possess the basic knowledge about auditing and that they can be a member of an audit team.
2. The **"Certified Auditor"** credential recognizes that individuals have the necessary knowledge to participate in an audit and that they possess the basic skills to conduct a management system certification audit having been members of an audit team.
3. The **"Certified Lead Auditor"** credential recognizes that individuals have mastered the audit techniques and demonstrate the audit competences to manage an audit team.
4. The **"Certified Senior Lead Auditor"** credential is targeted toward professionals who have extensive experience in auditing.

The main implementer credentials:

1. The **"Certified Provisional Implementer"**credential recognizes that individuals have the basic knowledge to participate in the implementation and management of a management system.
2. The **"Certified Implementer"**credential recognizes that individuals have the necessary knowledge to participate in the implementation and management of a management system.
3. The **"Certified Lead Implementer"**credential recognizes that individuals are equipped with the skills needed to implement a management system and possess the competences of managing a team.
4. The **"Certified Senior Lead Implementer"** credential is targeted toward professionals who have extensive experience in implementation projects.

The **"Master"** credential recognizes that individuals have mastered the basic concepts, approaches, methods, and techniques to lead an audit team and to lead the implementation of a management system.

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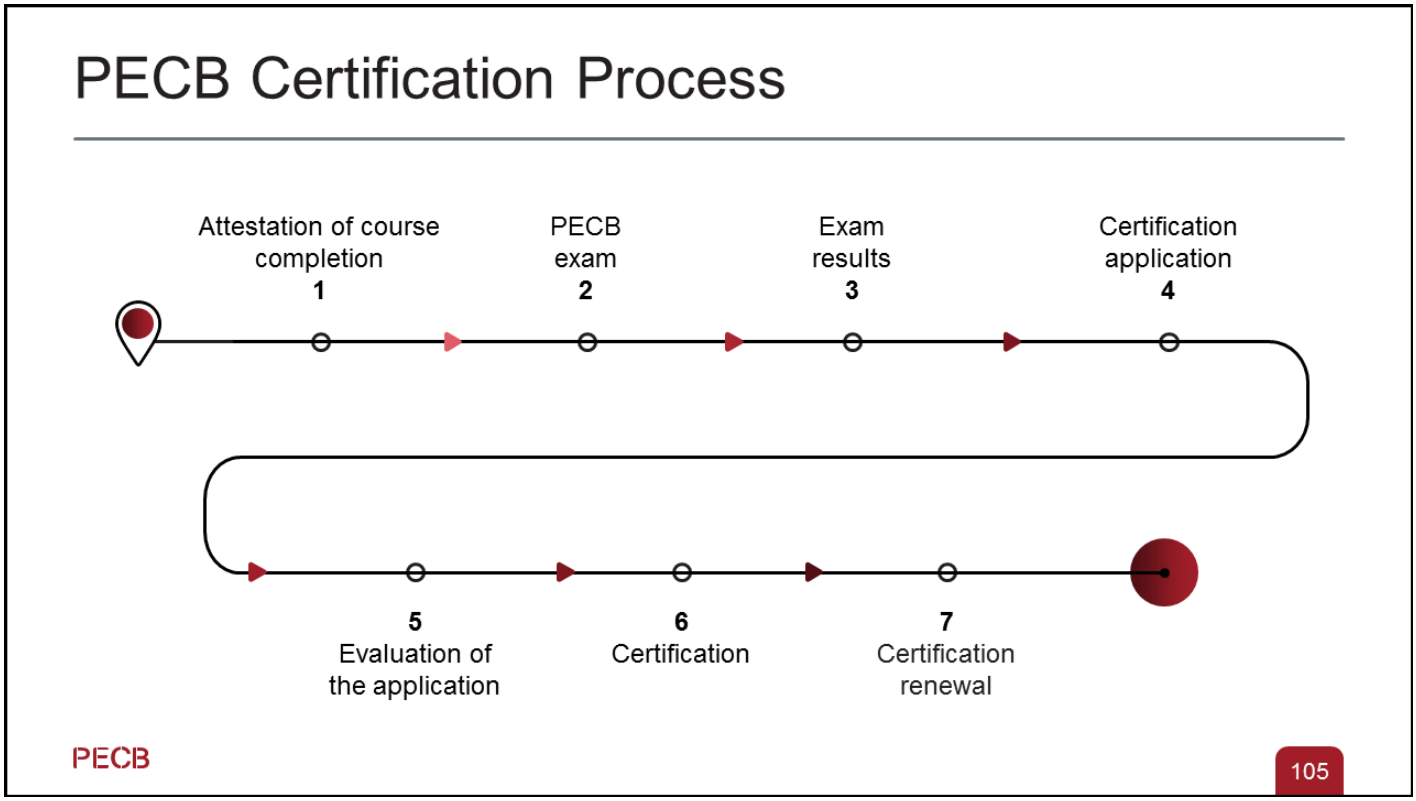

Passing the exam is not the only prerequisite to obtain the "PECB Certified ISO/IEC 27001 Lead Implementer" credential. The professional experience records will also be taken into account. If candidates have successfully passed the exam but do not have the required level of experience, they will not be able to claim the "PECB Certified ISO/IEC 27001 Lead Implementer" credential.

Important note: Candidates who have completed the training course with one of our partners, are provided with a coupon code that includes two exam attempts (first take and retake), certification application and the first year of Annual Maintenance Fee (AMF). This coupon code is valid within a 12-month period from the receive date.

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1. Attestation of Course Completion

Continuing Professional Development credits

PECB BEYOND RECOGNITION

Attestation of Course Completion

This document confirms that

John Doe

Has completed the

ISO/IEC 27001 Lead Implementer Training Course

This attestation of course completion is worth 31 Continuing Professional Development (CPD*) credits.

City/Country Location: XX, XX
Course Completion Date: 2024-04-25

This document is not a PECB Professional Certificate.

*1 CPD credit is worth 1 hour of activity. For more information, consult the PECB CPD policy.

CPD CERTIFIED The CPD Certification Service

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After attending the training course and submitting the **Training Course Evaluation Form**, an Attestation of Course Completion will be generated at the candidate's myPECB Dashboard under the **My Courses** tab. The Attestation of Course Completion is worth 31 CPD (Continuing Professional Development) credits.

Note: It is important to not confuse the Attestation of Course Completion with the actual certificate. The former is only a confirmation of having participated a training course, not gaining a certificate. To obtain a certificate, candidates will have to pass the exam, apply for certification, and get certified once the evaluation of the application is approved.

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2. PECB Exam

- The objective of the certification exam is to ensure that candidates have understood and mastered the implementation of an ISMS based on ISO/IEC 27001.
- The exam for this training course is going to be in the multiple-choice format.
- For specific information about exam types, languages available, and other details, please visit the PECB website at https://pecb.com.
- For more information about the examination process, please visit https://pecb.com/examination-rules-and-policies.

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- The PECB Examination Committee ensures that the development and adequacy of the exam questions are maintained based upon current professional practice.
- To take an exam in a particular language, please ask the trainer or contact us by sending an email toexamination@pecb.com.
- All competency domains are covered in the exam. To read a detailed description of each competency domain, please visit the PECB website athttps://pecb.com.

For **multiple-choice,open-book** exams, candidates are allowed to use the following reference materials:

- A hard copy of the main standard
- Training course materials (accessed through PECB Exams app and/or printed)
- Any personal notes taken during the training course (accessed through PECB Exams app and/or printed)
- A hard copy dictionary

The multiple-choice, open-book exams include both stand-alone and scenario-based questions. These questions consist of a stem and three options used to evaluate the competency of a candidate in a multiple-choice exam. Only one of the options is correct.

What is the difference between stand-alone and scenario-based questions?

Stand-alone questions stand independently within the exam and are not context-depended, whereas scenario-based questions are context-dependent, i.e., they are developed based on a scenario which a candidate is asked to read and is expected to provide answers to one or more questions related to that scenario.

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3. Exam Results

There are two possible exam results:

PASS

- Candidates will receive an exam number via email to apply for their certification.

FAIL

- Candidates who fail the first exam attempt are eligible to retake for free the exam within a 12-month period from the date the coupon code is received.
- For paper-based exams, candidates should contact the exam provider to determine the exam retake date. For online exams, candidates can use the initial coupon code to schedule the exam directly on the website.

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Exams are reviewed by qualified examiners who are assigned anonymously.

To ensure independence and impartiality and to avoid conflicts of interest, trainers, training course organizers, and invigilators do not participate in the exam review process or the certification process.

For candidates who fail the exam, a list of the domains where they have performed poorly will be added to the email to help them prepare better for a retake.

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4. Certification Application

General process

- After successfully passing the exam, candidates can apply online to get their PECB certification at https://pecb.com.
- For detailed information on the certification process, please go to https://pecb.com/pecb-certification-process
- For more information about the certification rules and policies, please visit https://pecb.com/certification-rules-and-policies.

When applying, candidates must provide the following information:

- Their contact details
- Their professional and audit experience records
- At least two references

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After successfully passing the exam, candidates have a maximum period of three years to submit a professional file in order to obtain a professional credential related to the ISO/IEC 27001 certification scheme. Candidates may apply at the same time for more than one professional credential related to the ISO/IEC 27001 certification scheme (e.g., Lead Implementer, Senior Lead Implementer, or Master) if all requirements are met.

When applying, candidates must provide the following information:

1. **Their contact details**
   - Candidates should write their name as they wish it to appear on their certificate (in ASCII format). Before submitting their certificate application, candidates should make sure to review the accuracy of the contact details they have provided when creating their PECB account. The certificate will be issued with the name that they provided when they created the account. To update the name in their PECB account, candidates should contact customer@pecb.com
2. **Their professional and project experience records**
   - Candidates must provide a resume to present their professional experience. Work experience can be any activity showing that they have skills and general knowledge about the functioning of an organization.
   - For project experience, candidates should make sure to indicate the number of hours completed.
   - Educational degrees or the like do not replace work experience.
3. **At least two references**
   - References (colleagues, partners, supervisors, etc.) that candidates provide must confirm their experience. It is important that the references (those individuals) know the candidate enough to prove their qualifications.
   - The candidate's application will be assessed once the references have been submitted.
   - **Note:** Providing references is not required for credentials such as Foundation, Transition, and Provisional.

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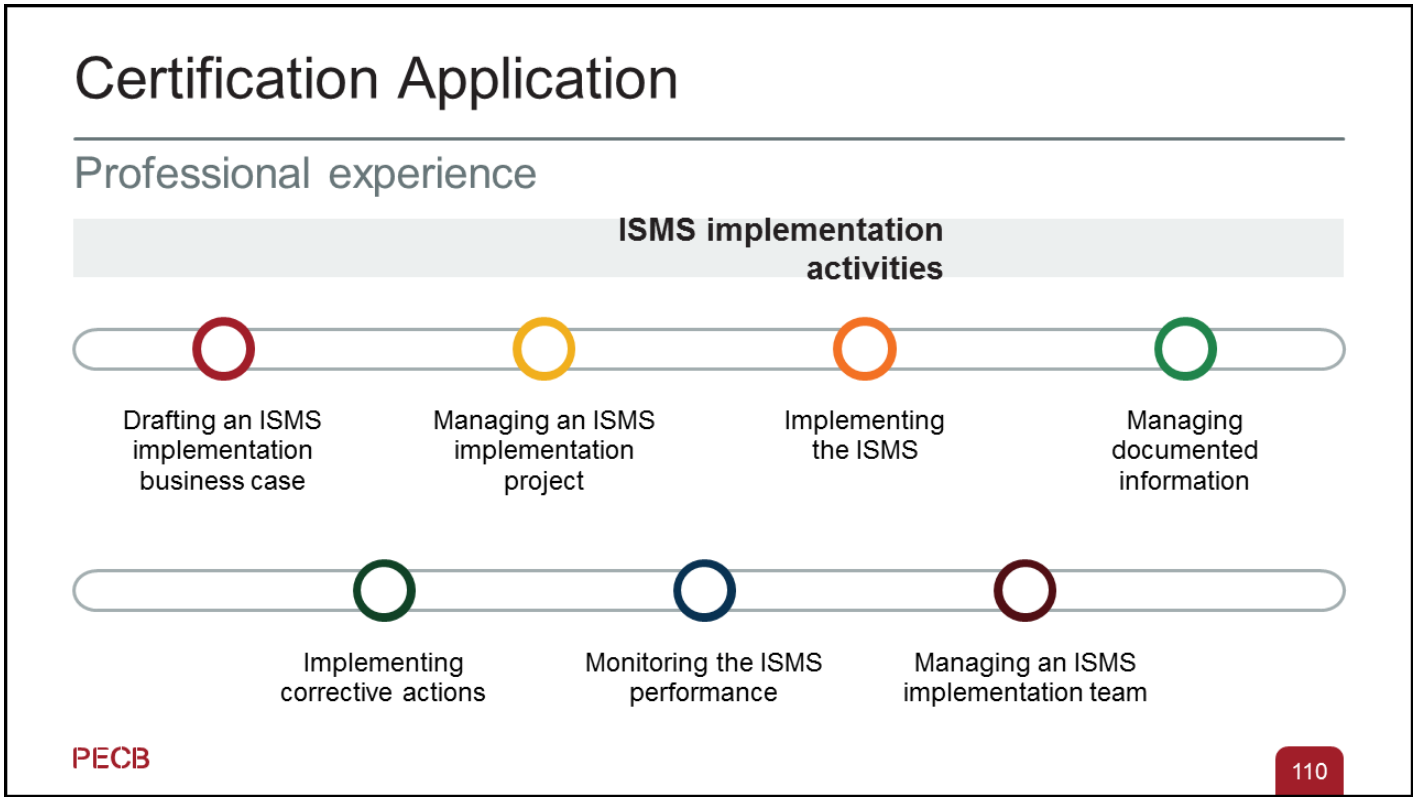

For example, a consultant who has conducted a risk assessment for a client to accompany the implementation of its compliance framework will be considered as having relevant experience.

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5. Evaluation of the Application

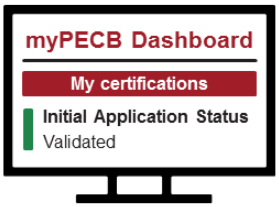
Once the application is complete, PECB will conduct an evaluation of the candidates' application.

Candidates' references will be contacted to validate:

- Their work experience
- Their personal and professional attitude

Candidates' application will not be evaluated until their references have been submitted.

Candidates can validate if their references have been submitted within their **PECB Account** Dashboard under the **My Certifications** tab.



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References will be contacted to complete a short questionnaire in order to attest to candidates' experience and evaluate their personal and professional qualities (according to the 13 Professional Behavioral Skills defined by ISO 19011).

Candidates can validate if their references have been submitted within their PECB account under the **My Certifications** tab. If their respondents are late, candidates should follow up with them to ensure that they have received the reference request.

In case PECB is unable to contact one of the references or the questionnaires were not answered, candidates will be asked to provide further references.

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6. Certification

- Once the application of candidates is approved, PECB is going to issue a professional certificate in PDF format which can be downloaded from their PECB account.
- The certificate contains the certification number which candidates can validate on the PECB website, https://pecb.com, by following the tab “Certification Verification.”
- Only candidates that fulfill all the criteria for certification can hold the “PECB Certified ISO/IEC 27001 Lead Implementer” credential.
- PECB has partnered with Credly to offer you the chance to claim a digital badge. You can share the badges online safely and easily. For more information, please go to https://pecb.com/pecb-digital-badges.

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When candidates are certified, they receive a notification from the system to download the certificate from their PECB account.

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7. Certification Renewal

How to renew the certification?

PECB Certifications are valid for three years. In order to maintain and renew a certification, PECB certified professionals must:

- Submit CPDs (Continuing Professional Development)
- Pay AMFs (Annual Maintenance Fee)
- Adhere to the PECB Code of Ethics

To renew their certification, PECB certified professionals must demonstrate sufficient hours of CPD activities related to the certification scheme. It is not mandatory to fulfill the required hours of CPD activities each year; they can instead be fulfilled within the three-year certification cycle.

In addition to CPDs, PECB Certified professionals will have to pay the AMFs (discussed on the following slides).

PECB certification(s) can be renewed online through the PECB Dashboard, by logging into the dashboard (https://pecb.com/en/login), clicking **My Certifications**, and then the **Renew** button. If after three years the recertification requirements are met, the certification will be renewed.

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Continuing Professional Development

- Continuing Professional Development (CPD) is a portfolio structure for demonstrating, documenting, and tracking the skills, knowledge, and experience acquired by professionals after their initial certification. CPDs are important for updating professional experience, acknowledging achievements, and demonstrating professional activities conducted.
- PECB certified professionals will need to provide PECB with the required hours of implementation-related tasks they have performed and/or other CPD activities that are considered eligible. CPDs can be submitted at any time, by logging into your PECB Dashboard, and clicking on ***My Certifications > CPD Info > Submit CPD.***

Certification	Annual CPD hours	3-year/Total CPD hours	Activities
Implementer	20 hours	60 hours	Hours of project experience, implementation, or consulting-related tasks, training, private study, coaching, attendance of seminars and conferences, or other relevant activities
Lead Implementer	30 hours	90 hours	
Senior Lead Implementer	60 hours	180 hours	

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To support certified professionals earn CPD credits, PECB continually organizes webinar sessions, provides opportunities for writing articles, participating in trainings and events, and more.

To learn more about CPDs obtained with PECB or other means, the CPD categories, their descriptions, the evidence required, the number of CPD credits awarded, and how to calculate CPDs, please visit our published CPD policy at https://pecb.com/pdf/brochures/cpd-policy.pdf.

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Annual Maintenance Fee

A PECB certification requires the payment of Annual Maintenance Fees (AMF). The annual reporting begins with the initial certification date; however, the maintenance fee for the first year is included in the certification application payment.

The annual maintenance fee can be paid online through your PECB dashboard, clicking ***My Certifications***, and then the ***Submit AMF*** button. If CPDs have not been submitted, certified professionals will be required to do so before the AMF payment.

For the last year of the certification cycle, the button to pay the AMF will be changed from ***Submit AMF*** to ***Renew***.

Certification	AMF (rate per year)
Foundation, Provisional, and Transition	None
All other certifications	$120

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For example, if a certification was issued on 2019-01-15 and is valid until 2022-01-15, the following requirements apply:

- **First AMF:** 2019-01-15 until 2020-01-15 – No payment of AMF required
- **Second AMF:** 2020-01-15 until 2021-01-15 – Payment of AMF required
- **Third AMF:** 2021-01-15 until 2022-01-15 – Payment of AMF required

If the certified professional fails to fulfill either of these CPD or AMF requirements, the certification will be downgraded.

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PECB Certification of Persons

- A personal certification is a formal recognition by PECB which states that the individual has proficiency within, and a comprehension of, a specified body of knowledge.
- Individuals can apply for various professional credentials in the PECB certification schemes. Each PECB certification has specific education and experience requirements.

Example:

PECB Certified ISO/IEC 27001 Lead Implementer

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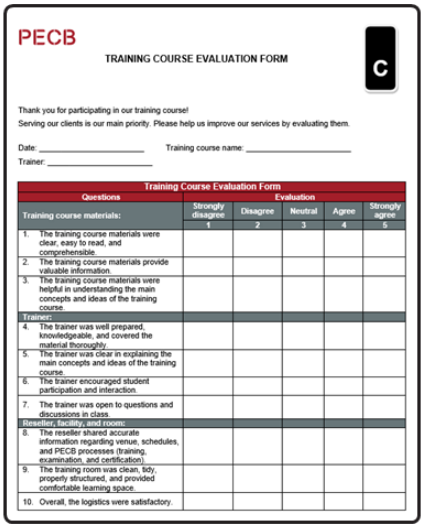
Evaluation of the Training Course

Training course evaluation form



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TRAINING COURSE EVALUATION FORM

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Thank you for participating in our training course!
Serving our clients is our main priority. Please help us improve our services by evaluating them.

Date: _____ Training course name: _____
Trainer: _____

Training Course Evaluation Form					
Questions	Evaluation				
Training course materials:	Strongly disagree 1	Disagree 2	Neutral 3	Agree 4	Strongly agree 5
1. The training course materials were clear, easy to read, and comprehensible.					
2. The training course materials provide valuable information.					
3. The training course materials were helpful in understanding the main concepts and ideas of the training course.					
Trainer:					
4. The trainer was well prepared, knowledgeable, and covered the material thoroughly.					
5. The trainer was clear in explaining the main concepts and ideas of the training course.					
6. The trainer encouraged student participation and interaction.					
7. The trainer was open to questions and discussions in class.					
Reseller, facility, and room:					
8. The reseller shared accurate information regarding venue, schedules, and PECB processes (training, examination, and certification).					
9. The training room was clean, tidy, properly structured, and provided comfortable learning space.					
10. Overall, the logistics were satisfactory.					

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Under your**myPECB Dashboard,**at the**My Courses tab,**you will find the training courses you have completed and are currently enrolled in. You will be asked to fill in the Training Course Evaluation Form when you finish the training course. You can submit your evaluation by clicking the Training Evaluation link. After submitting the training course evaluation form, an Attestation of Course Completion will be available to download.

Another way of providing feedback for training course materials is via the Kate application.

We strive to constantly improve the quality and the practical relevance of our trainings. Therefore, your opinion on this training is of great value to us.

We would be very grateful if you could provide us with your evaluation of the training course and the trainer(s).

Moreover, if you have any suggestions for improving PECB's training course materials, we would like to hear from you. Please open a ticket directed to the Training Development Department on PECB's website (https://pecb.com) in the**Contact Us**section. We thoroughly read and evaluate the input we get from our members.

In case of dissatisfaction with the training (trainer, training room, equipment, etc.), the examination, or the certification processes, please open a ticket under the **Make a complaint** category on PECB's website (https://pecb.com) in the **Contact Us** section.

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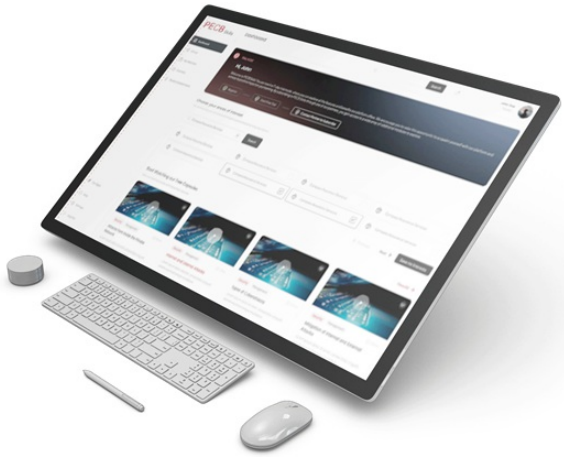
PECB Skills



PECB Skills is a new online learning platform carefully designed to bring development strategies, processes, and tools into your business through engaging micro lessons from highly experienced professionals.

www.mypecb.com

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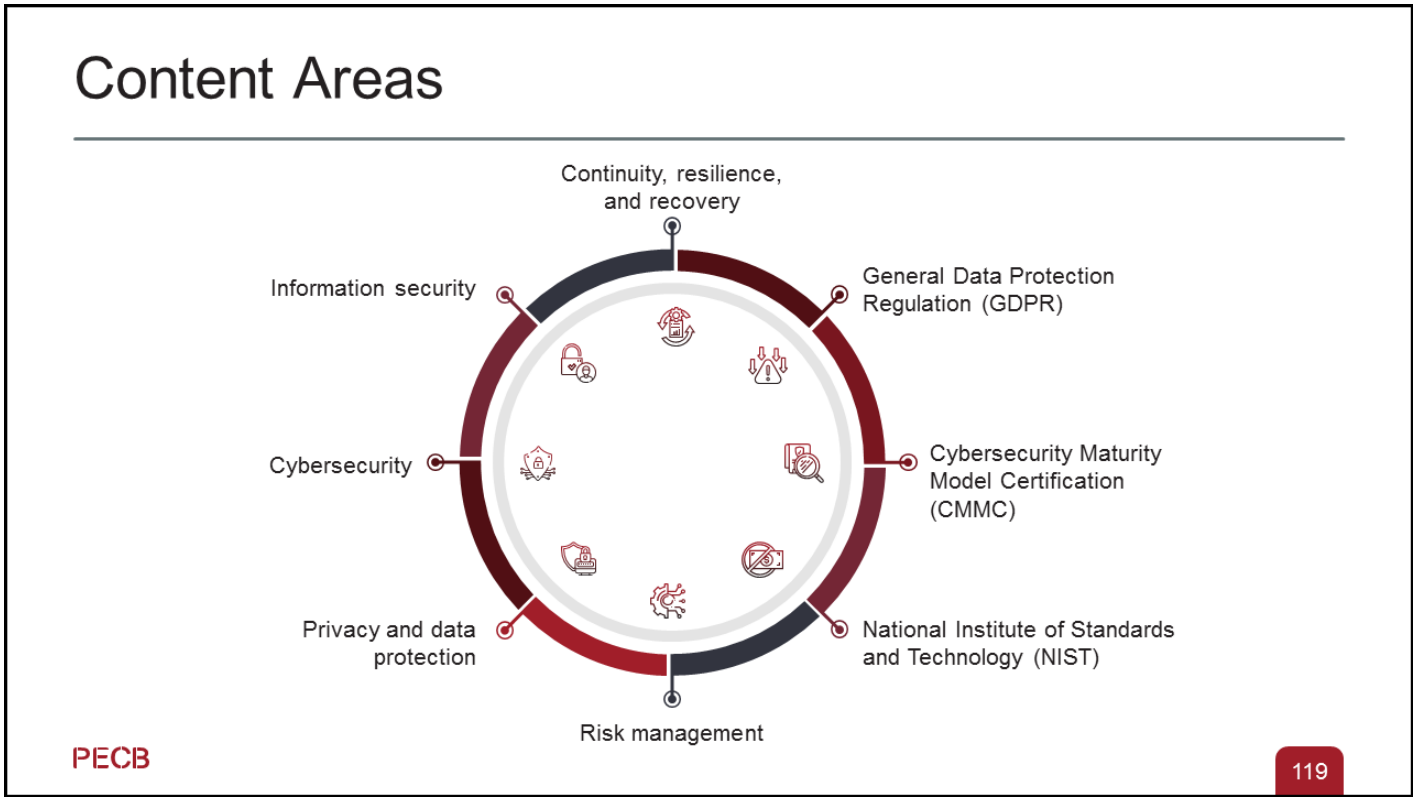

PECB Skills offers interactive micro lessons for all core content areas and more, providing professionals with a wide range of choices.

- **Continuity, resilience, and recovery:** Learn to build robust systems, ensuring business continuity even in unforeseen circumstances
- **Information security:** Equip yourself with the skills to protect vital data and maintain the integrity of information systems
- **Cybersecurity:** Stay vigilant and ahead of cyber threats with best practices and advanced techniques
- **Privacy and data protection:** Delve into the critical aspects of safeguarding personal data and ensuring privacy in an interconnected world
- **Risk management:** Navigate through uncertainties with confidence, identifying and mitigating potential risks
- **National Institute of Standards and Technology (NIST):** Engage with guidelines and standards for ensuring cybersecurity and privacy, backed by one of the most reputable institutions in the field
- **Cybersecurity Maturity Model Certification (CMMC):** Familiarize yourself with the defense standard for ensuring cybersecurity throughout the Defense Industrial Base (DIB)
- **General Data Protection Regulation (GDPR):** Master the European Union's benchmark regulation on data protection and privacy

With PECB Skills, you're not just learning; you are future-proofing your career and making strides in the dynamic digital ecosystem. For more information, please visit www.mypecb.com or contact us at marketing@pecb.com.

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Certification

PECB Skills Certificates

Elevate your profile and unlock new opportunities through these recognitions which will further enhance your professional standing and growth:

- **Attestation of Module Completion:** Demonstrate proficiency in specific modules and receive 1 CPD credit.
- **Essentials Certificate:** Demonstrate your competencies through a comprehensive 4-hour program, earning 4 CPD credits.

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Our micro-courses are accredited by the ANSI National Accreditation Board (ANAB), a testament to their quality and adherence to rigorous educational standards.

Note: Modules comprise four short video capsules.

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Other PECB Training Courses and Certifications

ISO/IEC 27001 Lead Auditor (five days)

- Fundamental principles and concepts of an information security management system (ISMS)
- Fundamental audit concepts and principles
- Preparing for an ISO/IEC 27001 audit
- Conducting an ISO/IEC 27001 audit
- Closing an ISO/IEC 27001 audit
- Managing an ISO/IEC 27001 audit program

ISO/IEC 27005 Lead Risk Manager (five days)

- Fundamental principles and concepts of information security risk management
- Implementation of an information security risk management program
- Information security risk assessment
- Information security risk treatment
- Information security risk communication, monitoring, and improvement
- Information security risk assessment methodologies

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PECB Certified ISO/IEC 27001 Lead Auditor (five days)

The ISO/IEC 27001 Lead Auditor training course enables participants to develop the necessary ability to perform an information security management system (ISMS) audit by applying widely recognized audit principles, procedures, and techniques. During this training course, participants acquire the knowledge and skills to plan and carry out internal and external audits in compliance with ISO 19011 and the certification process according to ISO/IEC 17021-1.

PECB Certified ISO/IEC 27005 Lead Risk Manager (five days)

The ISO/IEC 27005 Lead Risk Manager training course enables participants to become proficient in the fundamentals of information security risk management, including the planning of a risk management program, analysis, evaluation, risk treatment, risk communication, and surveillance. Through a variety of activities, exercises based on real-life cases, discussions, and demonstrations with risk modeling tools, participants will be able to perform an optimal risk evaluation. The participants will also gain a thorough understanding of risk assessment methods, such as OCTAVE, EBIOS, MEHARI, CRAMM, and harmonized TRA. This training course corresponds with the implementation process of the ISMS framework presented in ISO/IEC 27001.

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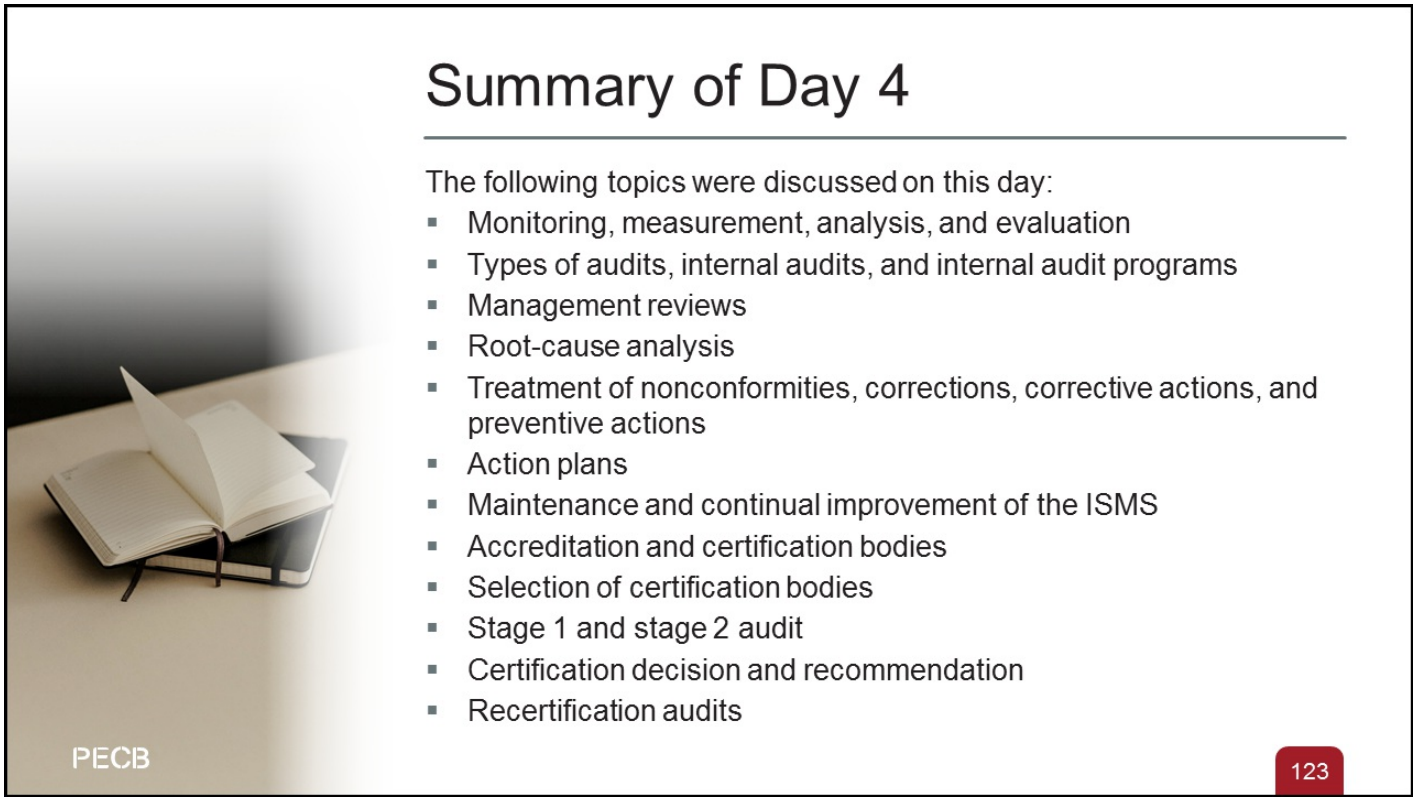

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Homework 11–12 (optional)

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Note: To complete Homework 11-12, please go to the Exercises Sheet.

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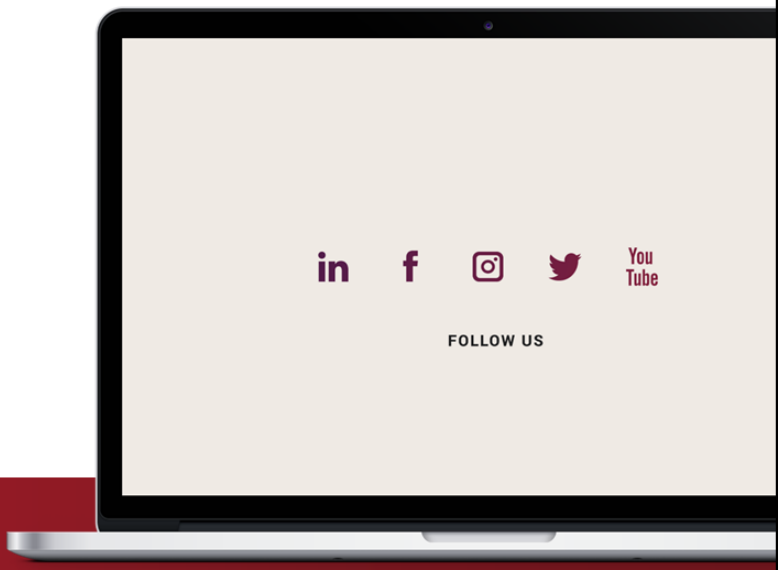

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